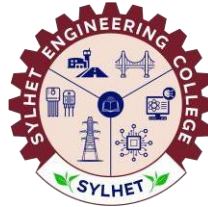


SYLHET ENGINEERING COLLEGE, SYLHET

Department of Computer Science & Engineering

CSE- 800



Bachelor's degree certificate generation and Verification using ZKProofs

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This is to certify that the work presented in this thesis entitled, “**CertiChain: A Decentralized, Gas-Optimized Academic Credential Issuance, Zero-Knowledge Verification, and Multi-Tier Governance Platform on Polygon Blockchain**”, is the outcome of the research carried out by the authors under the supervision of **Md. Rasel**, Lecturer, Department of Computer Science and Engineering, Sylhet Engineering College, Sylhet-3100, Bangladesh.

It is also declared that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma, or other qualification.

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ABSTRACT

Academic credentials represent a cornerstone of societal trust, validating educational qualifications for employment, higher studies, professional licensing, and immigration. However, traditional credential verification mechanisms remain plagued by vulnerabilities in paper-based documents, centralized databases as single points of failure, administrative verification bottlenecks, and severe privacy violations resulting from over-disclosure of sensitive transcripts [1, 2, 6]. This thesis presents CertiChain, an enterprise-grade, decentralized academic credential issuance, zero-knowledge verification, and multi-tier institutional governance platform deployed on the Polygon blockchain ecosystem [3, 4]. CertiChain establishes a comprehensive 5-portal role-based architecture [1, 2, 10]: Student Portal (featuring 2FA registration via OTP, dynamic credential access, printable/shareable degree certificates and semester grade sheets, document vault, and request center for academic record corrections, duplicate reissues, and Medium of Instruction [MOI] / Sealed Envelope services); Department Minter Portal (equipped with bulk CSV/XLSX and manual ingestion, legacy graduate archiving, session-based student dossiers, batch/provisional certificate generation with dual audit ledgers, single-student transcript builders, and multi-category request resolution); Signatory Portal (introducing a multi-signature review queue, cryptographic endorsement history, and digital signature authoring); Academic Admin Controller Portal (facilitating centralized cohort degree minting, RBAC staff authorization, cohort result publication, certificate revocation with atomic on chain burning and re-minting, system-wide notices, comprehensive multi-category audit trails, and institutional branding); and Public Verification Portal (enabling multi-modal instant verification via cryptographic Keccak-256 hash/code, live camera/image QR scanning, PDF upload, and client side privacy-preserving zk-SNARK proof verification and biometric face matching) [5, 8, 9]. By implementing a hybrid on-chain/off-chain architecture—anchoring only 32-byte cryptographic commitments on Polygon Amoy while offloading bulky PDFs and metadata to IPFS/Pinata and maintaining relational telemetry via Prisma ORM with SQLite—CertiChain minimizes on-chain gas costs by over 99% compared to Ethereum mainnet [4, 8, 10]. Furthermore, the integration of zk SNARKs enables selective qualification proofs (e.g., proving CGPA $\geq$ threshold, degree completion, departmental affiliation, passing year) without disclosing complete student transcripts [4, 5, 9]. Formal evaluation demonstrates sub-second verification latencies, complete resistance against certificate forgery and unauthorized tampering, and full operational transparency across the entire credential lifecycle [1, 3, 6].

Keywords: Blockchain, Academic Credentials, Polygon, IPFS, Smart Contracts, Keccak-256, zk SNARK, Zero-Knowledge Proofs, Multi-Tier Governance, Role-Based Access Control (RBAC), Biometric Verification, 5-Portal Architecture.

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List of Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
CID	Content Identifier
CGPA	Cumulative Grade Point Average
DApp	Decentralized Application
IPFS	InterPlanetary File System
NFT	Non-Fungible Token
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
SRS	Software Requirements Specification
zk-SNARK	Zero-Knowledge Succinct Non-Interactive Argument of Knowledge
ZKP	Zero-Knowledge Proof
SEC	Sylhet Engineering College
SUST	Shahjalal University of Science and Technology
OTP	One-Time Password
QR	Quick Response
RPC	Remote Procedure Call
ORM	Object-Relational Mapping

Chapter 1

Introduction

Academic certificates are among the most important documents issued by an educational institution. They are used when graduates apply for employment, higher education, professional licensing, scholarships, immigration, and other opportunities. Because a certificate represents a claim about a person's academic achievement, the institution and the external verifier both need confidence that the document is genuine and has not been altered.

In a conventional environment, certificate management is usually centered on institutional databases and physical or PDF documents. A graduate may keep a paper certificate for years, while an employer or another institution may need to contact the issuing authority to confirm its authenticity. This approach can work at a small scale, but it becomes inconvenient when verification requests increase, records are distributed across different offices, or certificates need to be verified internationally.

CertiChain addresses this problem by treating a certificate as a verifiable digital credential rather than merely a file. The system creates a cryptographic representation of standardized certificate data, anchors that representation to Polygon, stores larger files through IPFS, and provides dedicated interfaces for students, departmental issuers, institutional administrators, and public verifiers.

1.1 Problem Definition

- **Certificate Forgery:** Forgery and alteration of certificates are difficult to detect reliably using paper documents or ordinary PDFs.
- **Manual Verification:** Manual verification may require communication with the issuing institution and can take substantial time.
- **Centralized Storage:** A centralized database can become a single point of failure and may be exposed to unauthorized modification.
- **Privacy Risk:** Publishing complete academic records for public verification creates privacy risks because verification does not always require all personal information.

- **Blockchain Storage Cost:** Large files such as certificate PDFs, photographs, and supporting documents are inefficient to place directly on a blockchain.
- **Role Management:** Certificate corrections require a controlled institutional process; simply overwriting an old record would weaken auditability.
- **Privacy-Preserving Verification:** Different stakeholders need different permissions, but traditional systems may not clearly separate student, departmental, and central administrative responsibilities.

1.2 Motivation

The motivation behind CertiChain is not to replace every existing academic information system with blockchain. Instead, the goal is to use blockchain where it provides the greatest value: integrity, non-repudiation, public verification, and an auditable history of important credential events. The design also recognizes the limitations of blockchain storage. Academic records can be large and contain sensitive information, so the system deliberately keeps detailed operational and document data off-chain and stores only compact commitments on-chain.

Privacy is another major motivation. A graduate applying for a job may need to prove that the CGPA is above a minimum threshold, but the employer may not need to know the exact CGPA, roll number, date of birth, or complete transcript. The zero-knowledge module is therefore designed around selective claims rather than full disclosure.

1.3 Aim of the Study

The main aim of this project is to design and implement a secure, privacy-aware, gas-efficient, and institutionally governed academic credential issuance and verification platform using Polygon blockchain, decentralized storage, cryptographic hashing, zero-knowledge proofs, and client-side biometric verification.

1.4 Objectives of the Study

The main objective of this project is to develop a secure, decentralized, and privacy-aware academic credential platform that can simplify the process of certificate issuance, management, and verification. The proposed system aims to use blockchain technology, decentralized storage, and privacy-preserving verification techniques to improve certificate

integrity and reduce the risk of fraud and unauthorized modification. The specific objectives of the project are as follows:

- **Blockchain-Based Certificate Registry:** The system creates a tamper-resistant digital ledger of academic certificates using blockchain. This ensures that credentials can be independently verified while making unauthorized modifications virtually impossible.
- **Keccak-256 Data Integrity:** Standardized certificate data is mathematically converted into a unique cryptographic hash. Any minor alteration to the original document instantly changes the resulting hash, allowing the system to easily detect tampering.
- **Off-Chain Document Storage:** To optimize performance, heavy files like PDFs and high-resolution photographs are stored off-chain. Cryptographic commitments permanently link these files to the blockchain, ensuring their integrity without overloading the network.
- **Multi-Method Public Verification:** The platform accelerates the authentication process by offering various instant verification methods. Verifiers can easily check credentials using QR codes, live camera scans, serial numbers, or cryptographic proofs.
- **zk-SNARK Privacy Preservation:** Zero-knowledge proofs (zk-SNARKs) empower students to validate specific academic claims—such as meeting a minimum CGPA requirement—without exposing their complete transcript or sensitive personal information.
- **Role-Based Access Portals:** The system features distinct, secure interfaces tailored for students, minters, administrators, signators and verifiers. This strict role-based architecture ensures that users can only access functions relevant to their specific clearance level.
- **Certificate Lifecycle Management:** The platform governs the entire lifecycle of a credential, enabling secure corrections, on-chain revocations, and re-minting. This institutional workflow guarantees a transparent and permanent audit trail rather than simply overwriting data.
- **Gas Cost & Storage Minimization:** By anchoring only compact cryptographic hashes on-chain and shifting the bulk of the data off-chain, the system drastically reduces transaction gas fees and prevents blockchain network bloat.
- **User-Friendly Web Interface:** The system abstracts complex Web3 and blockchain mechanics behind an intuitive graphical interface. This guarantees seamless usability for all stakeholders, even those with zero technical blockchain knowledge.

- **Comprehensive System Testing:** The developed platform undergoes rigorous functional, security, performance, and usability testing. This comprehensive evaluation ensures the architecture meets its intended operational requirements and remains highly resilient.

1.5 Research Questions

- Q1.** How can blockchain be used to provide independently verifiable academic credentials without storing large sensitive documents directly on-chain?
- Q2.** How can zero-knowledge proofs reduce unnecessary disclosure during academic qualification verification?
- Q3.** How can institutional roles and approval stages reduce the risk of unauthorized credential issuance or modification?
- Q4.** What design choices can reduce gas consumption while preserving certificate integrity and auditability?
- Q5.** Can the proposed multi-portal architecture provide a practical and usable workflow for students, certificate issuers, administrators, and public verifiers?

1.6 Scope of the Project

The current project focuses on academic credential management for Sylhet Engineering College and is designed around degree credentials, academic results, transcripts, student profiles, and verification workflows. The implementation supports a public verifier, student portal, department mint terminal, and central academic admin controller. The system is designed for Polygon Amoy testnet in the current project specification.

- certificate issuance, certificate verification, QR verification, student credential access, academic result display, IPFS-backed documents, role-based governance, correction requests, revocation, burn and re-mint, zk-proof generation/verification, and client-side face comparison.
- database persistence using Prisma ORM with SQLite for operational and institutional records.

- smart contract interaction through wagmi/viem and wallet-based Web3 connectivity.
- Excluded from the current implementation: full production deployment on Polygon mainnet, institutional-wide integration with legacy SIS platforms, and universal interoperability with every external credential standard.

1.7 Research Contribution, Originality and Novelty

The contribution of this project lies in combining several mechanisms that are often treated separately: compact blockchain commitments, decentralized document storage, selective zero-knowledge verification, client-side biometric checking, and multi-tier institutional governance. The system does not simply place a certificate on a blockchain. It defines how a credential moves through its complete lifecycle—from registration and issuance to verification, correction, revocation, burning, and re-issuance.

- A gas-aware hybrid storage architecture that stores only a 32-byte certificate hash on-chain while retaining large files off-chain.
- A multi-actor governance model separating student requests, departmental endorsement, and central administrative execution.
- Privacy-preserving verification presets for minimum CGPA, degree completion, departmental affiliation, and passing year.
- A public verification workflow that combines cryptographic validation with optional client-side face comparison.
- A deterministic certificate correction process that creates a new certificate identity rather than silently overwriting historical information.
- A unified institutional platform that brings together academic credentials, semester results, transcripts, requests, notifications, audit records, and verification.

Chapter 2

Literature Review

This chapter presents a comprehensive review of existing literature on blockchain-based academic certificate issuance and verification systems, with particular emphasis on certificate security, decentralized verification, blockchain-based authentication, QR-code verification, and decentralized storage using the InterPlanetary File System (IPFS). The review synthesizes findings from recent research studies published between 2023 and 2026, covering hybrid blockchain architectures, Ethereum-based certificate verification, trusted issuer validation, QR-code-based authentication, and permissioned blockchain solutions (Deepika et al., 2026; Jaafar et al., 2024; Kadam et al., 2025; Noorhizama et al., 2023; Priyadarshini et al., 2025; Rani et al., 2024; Rustemi et al., 2023; Said et al., 2023; Sree, 2025; Watane et al., 2026).

The application of blockchain technology for academic certificate verification has emerged as a promising approach to address the limitations of conventional certificate management systems. Traditional certificate verification often depends on centralized institutional databases and manual communication between certificate holders, employers, and educational institutions. Such processes can be time-consuming and may introduce security and authenticity concerns. Watane et al. (2026) proposed Hy-Cert, a hybrid certificate verification system that utilizes blockchain to improve the security and reliability of certificate verification. Similarly, Deepika et al. (2026) investigated a blockchain-based certificate verification system and demonstrated the potential of blockchain technology for improving the authenticity and security of academic credentials.

A major advantage of blockchain-based certificate systems is the ability to maintain a tamper-resistant record of credential information. Priyadarshini et al. (2025) proposed a faster and integrated certificate authentication and issuer validation system based on blockchain. Their work emphasizes the importance of verifying not only the authenticity of a certificate but also the legitimacy of its issuing authority. This approach is particularly important in academic environments, where a verifier needs confidence that a credential was genuinely issued by an authorized educational institution.

Ethereum has also been widely investigated as a platform for blockchain-based academic credential management. Sree (2025) proposed a decentralized certificate issuance and

verification system using Ethereum blockchain technology. The study demonstrates how smart contracts can be used to support certificate issuance and verification without relying entirely on conventional centralized verification mechanisms. Ethereum-based approaches provide programmable verification logic and enable credential-related information to be associated with blockchain transactions.

Another important area of research focuses on the authenticity and validation of digital certificates. Kadam et al. (2025) introduced ValidCertify, a blockchain-based certificate authentication and verification scheme. Their research highlights the potential of blockchain to strengthen certificate authentication by providing a trustworthy digital verification mechanism. Such approaches demonstrate that cryptographic and blockchain-based validation can reduce dependence on manual verification and improve confidence in digital academic credentials.

The increasing number of blockchain-based academic credential systems has also resulted in systematic evaluations of existing research. Rustemi et al. (2023) conducted a systematic literature review of blockchain-based systems for academic certificate verification. Their study identifies blockchain as an important technology for improving certificate authenticity, integrity, transparency, and verification. However, the literature also indicates that different systems make different architectural choices regarding blockchain networks, storage mechanisms, verification methods, and institutional governance.

The application of blockchain for educational certificate verification has also been investigated in developing-country contexts. Said et al. (2023) proposed a blockchain-based conceptual model to address educational certificate verification challenges in Tanzania. Their research demonstrates that blockchain can provide a more transparent and reliable mechanism for academic credential verification while reducing dependence on traditional verification procedures. This is particularly relevant for countries where manual certificate verification may create administrative delays and additional costs.

Decentralized storage is another important component of blockchain-based credential systems. Jaafar et al. (2024) developed an educational certificate verification system that combines Ethereum blockchain and IPFS to enhance certificate security and authenticity. The combination of blockchain and IPFS provides a practical architectural model in which blockchain can be used for verification and integrity while decentralized storage can be used to manage certificate-related digital files. This hybrid approach is particularly useful because storing large certificate documents directly on a blockchain can be inefficient.

QR-code-based verification has also been explored as a practical mechanism for simplifying certificate authentication. Noorhizama et al. (2023) proposed a system for Ph.D. certificate verification using QR codes on the Ethereum blockchain. QR codes provide a convenient bridge between physical or digital certificates and blockchain-based verification services. A verifier can scan the QR code associated with a certificate and use the encoded information to initiate the verification process. This approach can significantly reduce the effort required for manual certificate checking and improve the accessibility of blockchain-based verification.

Permissioned blockchain architectures have also been considered for academic certificate management. Rani et al. (2024) proposed Educert-Chain, a secure and notarized educational certificate authentication and verification system using permissioned blockchain. Their work demonstrates that blockchain-based credential systems do not necessarily need to rely exclusively on public blockchain networks. Permissioned blockchain can provide controlled participation and institutional governance, which may be suitable for educational organizations requiring restricted access to credential-related operations.

Overall, the reviewed studies demonstrate that blockchain can provide significant benefits for academic certificate management, including tamper resistance, decentralized verification, improved authenticity, issuer validation, and reduced dependence on manual verification (Deepika et al., 2026; Jaafar et al., 2024; Kadam et al., 2025; Noorhizama et al., 2023; Priyadarshini et al., 2025; Rani et al., 2024; Rustemi et al., 2023; Said et al., 2023; Sree, 2025; Watane et al., 2026). Ethereum-based systems demonstrate the feasibility of implementing certificate-related smart contracts (Jaafar et al., 2024; Noorhizama et al., 2023; Sree, 2025), while IPFS-based architectures provide an approach for managing digital certificate files outside the blockchain (Jaafar et al., 2024). QR-code mechanisms further improve the usability of verification systems by connecting physical or digital certificates with blockchain records (Noorhizama et al., 2023). At the same time, permissioned blockchain approaches provide greater institutional control over credential management (Rani et al.).

Despite these developments, the reviewed literature also reveals several areas that can be further explored. Existing studies often concentrate on individual aspects of certificate management, such as blockchain authentication, QR-based verification, decentralized storage, or issuer validation. A comprehensive institutional platform that integrates certificate issuance, academic record management, decentralized document storage, public verification, privacy-preserving verification, role-based governance, certificate correction, revocation, and re-

issuance within a single lifecycle remains an important area for further development. CertiChain addresses this research direction by combining blockchain-based certificate integrity, hybrid storage, institutional governance, and multiple verification mechanisms into an integrated academic credential management platform.

Chapter 3

Background Study

2.1 Blockchain Technology

Blockchain can be understood as a distributed ledger in which transactions or state changes are recorded in a sequence of blocks and protected by cryptographic mechanisms. Its value for academic credentials comes from the combination of integrity, auditability, and independent verification. Rather than asking an institution to manually confirm every certificate, a verifier can check whether a known cryptographic commitment exists and whether its status is still valid.

For CertiChain, blockchain is not treated as a conventional database replacement. The blockchain is used as the trusted integrity layer. It records the minimum information required to prove that a credential commitment exists, who issued it, when it was issued, and whether it has been revoked.

2.2 Polygon Amoy Testnet

The current project specification targets Polygon Amoy, a test network used for development and testing. The project summary identifies Chain ID 80002 for Amoy. Using a testnet makes it possible to test wallet interaction, contract deployment, transaction confirmation, verification, and governance flows without treating the prototype as a production credential infrastructure.

Polygon was selected because the project requires frequent credential-related transactions and therefore benefits from a network designed to significantly reduce transaction overhead compared with directly placing the workload on the Ethereum mainnet. Executing a smart contract transaction (such as minting a certificate) on the Ethereum mainnet incurs steep gas fees, typically ranging from \$5.00 to \$50.00 depending on network congestion. Converted to local currency, this would cost approximately 607 Tk to 6,075 Tk per certificate issuance (assuming an exchange rate of 1 USD $\approx$ 121.50 BDT). In stark contrast, Polygon's Layer-2 scaling architecture reduces this transaction cost by over 99%. A standard certificate minting transaction on Polygon costs only about \$0.01 to \$0.05, which is equivalent to roughly 1.20 Tk to 6.00 Tk.

This massive cost efficiency makes the system economically viable for institutions handling

thousands of student records. Furthermore, the architecture is intentionally network-aware so that a future production deployment on the Polygon Mainnet can be evaluated seamlessly, separated from the current academic testnet implementation.

2.3 Smart Contracts

Smart contracts are self-executing contracts where the terms of the agreement are directly written into code. These contracts automatically enforce rules and execute transactions when predetermined conditions are met, making them an essential component of blockchain applications. It's also a program deployed to a blockchain that exposes predefined functions and maintains persistent state. In CertiChain, the CertificateRegistry contract acts as the on-chain certificate registry. It stores the certificate hash, issuer, issue time, revocation status, and revocation count. Access modifiers distinguish the contract owner from authorized department issuers.

2.4 Keccak-256 Hashing

Hashing converts an input into a fixed-size digest. CertiChain constructs a deterministic string from standardized certificate attributes and applies Keccak-256. Because the output is 32 bytes, the commitment is compact enough to be stored efficiently in a blockchain storage slot. A verifier can reconstruct the standardized input and compare the resulting digest with the on-chain record.

```
certHash = keccak256(  
    abi.encodePacked(  
        studentName, "|",  
        rollNo, "|",  
        regNo, "|",  
        department, "|",  
        degree, "|",  
        cgpa, "|",  
        passingYear  
    )  
)
```

The important property is sensitivity to change. If one of the standardized fields changes, the resulting digest also changes. This makes the hash useful as an integrity fingerprint rather than as a replacement for the full certificate.

2.5 InterPlanetary File System (IPFS)

To overcome the prohibitive costs and storage limitations of public blockchains, the system leverages the InterPlanetary File System (IPFS) as a decentralized, peer-to-peer off-chain storage layer. Unlike conventional web servers that rely on location-based addressing (where files are found via a specific URL or IP address), IPFS utilizes content addressing. When a file is uploaded to the network, it is mathematically processed to generate a unique, immutable Content Identifier (CID). Because the CID is derived directly from the file's data, any modification to the document will fundamentally alter the resulting hash, inherently preventing undetected tampering.

In this project, heavy digital assets—such as certificate PDFs, high-resolution student photographs, and structured academic metadata—are securely pinned to the IPFS network via the Pinata Cloud service. Rather than storing these bulky files directly on the Polygon blockchain, the system only records the resulting 32-byte CID within the smart contract. This hybrid architecture ensures that large files are kept entirely out of the expensive smart contract storage layer, preventing network bloat and minimizing transaction gas fees. At the same time, it allows the application to reliably and permanently retrieve the exact, unaltered academic content from the decentralized network whenever verification is required.

2.6 Zero-Knowledge Proofs

A zero-knowledge proof allows one party to demonstrate that a statement is true without revealing the underlying private value. In academic verification, this is useful because the verifier often needs a yes/no qualification rather than a complete transcript.

CertiChain defines four proof presets: minimum CGPA, degree completion, departmental affiliation, and passing year. For example, a minimum CGPA proof can demonstrate that a student's CGPA satisfies a threshold without exposing the exact CGPA. The project summary describes a Groth16/Plonk-style zk-SNARK proof structure and a compact JSON payload containing proof elements and public signals.

2.7 Client-Side Biometric Face Verification

The system includes a client-side face comparison mechanism intended to reduce the risk of credential sharing or proxy verification. The student's official photograph is retrieved from the credential record, while the verifier can capture a live webcam frame. Facial descriptors are compared in the browser, and the system produces a similarity score.

Keeping the comparison on the client is important for privacy. The design described in the project summary states that biometric computation is performed in the browser using TensorFlow.js/WebGL and that facial biometric data is not stored or transmitted to an external server as part of the comparison workflow.

2.8 Role-Based Access Control

Role-based access control means that a user receives permissions according to an assigned role rather than receiving unrestricted access to every function. CertiChain uses four functional actor categories: **Public Verifier, Student, Department Minter, and Academic Admin Controller**. This separation reduces the chance that one compromised account can perform every sensitive operation.

2.9 Merkle Tree Commitments

For graduating cohorts, storing each credential commitment independently can still create repeated transaction overhead. The project therefore includes a Merkle-tree-oriented bulk strategy. Individual student hashes are organized into a tree and the root can act as a compact commitment to a large set of credentials. This creates a foundation for future batch issuance and large-cohort processing.

2.10 QR Code-Based Verification

Quick Response (QR) Code-Based Verification enables third parties to instantly validate academic credentials without requiring account registration, Web3 wallet software, or manual data entry. In higher education verification, this allows employers and embassies to authenticate documents on-the-go with zero operational friction.

To facilitate this process, CertiChain embeds a cryptographic QR code onto official certificate PDFs, enabling several instant automated checks:

- **On-Chain Existence & Issuer Validation:** Confirms the certificate exists on Polygon Amoy and verifies the cryptographic signature of an authorized SEC Department Minter or Controller.
- **Active Status Check:** Confirms the credential is valid and has not been revoked or burned.
- **Block Timestamp Audit:** Retrieves the exact block confirmation timestamp from the ledger.
- **Biometric & Visual Authentication:** Displays an anti-forgery watermarked certificate preview and triggers client-side facial verification against the stored IPFS photo.

Under the hood, the public verification portal (/verify) processes live camera scans or uploaded QR images to query the smart contract in real time. This provides instantaneous, tamper-proof verification within seconds without relying on manual institutional communication

2.11 Hybrid Storage Principle

The central design principle is to assign each type of data to the layer where it is most appropriate. Immutable credential commitments and lifecycle state belong on-chain. Large documents belong in IPFS. Operational records, requests, notifications, profile information, and application state belong in the relational database. This division keeps the system practical while preserving blockchain-backed integrity.

Layer	Primary role	Examples
Polygon	Integrity and lifecycle commitment	Certificate hash, issuer, timestamp, revocation state
IPFS / Pinata	Large and content-addressed files	Certificate PDF, photo, metadata JSON
SQLite / Prisma	Operational and institutional data	Students, grades, requests, notifications, audit records
Browser	Interactive verification and privacy-preserving computation	Face comparison, proof generation, wallet interaction

Chapter 4

Related Work and Research Gap

Blockchain-based academic credential systems have been explored for more than a decade. Existing approaches generally share a core idea: the institution issues a credential, the recipient holds or shares it, and a verifier checks its authenticity using a cryptographic record. Open standards such as Blockcerts demonstrate that blockchain can support issuing, viewing, and verifying official records. More recent verifiable credential standards emphasize machine-readable, cryptographically secure, and privacy-respecting credentials.

The pre-defense materials supplied for this project also review earlier blockchain certificate systems, Ethereum-based credential models, IPFS-backed storage, Polygon implementations, and zero-knowledge proof research. These works show that blockchain can improve integrity and verification speed, but they also reveal recurring design tensions: public transparency can conflict with privacy, on-chain storage can be expensive, and institutional correction workflows are often under-specified.

4.1 Blockchain Credential Platforms

Blockcerts is an important reference point because it treats academic and professional certificates as blockchain-based official records that can be issued and verified without depending on a single centralized verification service. Its recipient-centric approach is particularly relevant to CertiChain. However, CertiChain is designed around a specific institutional governance model and extends the workflow with department-level endorsement, central approval, certificate correction, burn and re-mint operations, zk-proof presets, and client-side biometric checking.

4.2 Ethereum-Based Credential Systems

Earlier Ethereum-oriented systems demonstrated the feasibility of storing certificate information and issuing credentials through smart contracts. Their main limitation for a resource-constrained academic environment is the cost of putting large amounts of data directly on-chain. CertiChain addresses this by storing a compact 32-byte commitment on-chain and moving large files to IPFS.

4.3 IPFS-Backed Credential Storage

Using IPFS for certificate files is a common architectural response to the cost and size limitations of blockchain storage. The important distinction is that IPFS is used for file availability and content addressing, while the blockchain provides the institutional integrity anchor. CertiChain follows this separation but additionally keeps operational records in a structured database.

4.4 Zero-Knowledge Verification

Zero-knowledge proof research has created a strong foundation for privacy-preserving academic verification. Instead of revealing a complete transcript, a holder can prove a statement about a transcript. CertiChain adopts this concept at the application level by defining practical proof presets that correspond to common recruitment and admissions questions.

4.5 Digital Certificates in Bangladesh

In Bangladesh, academic credential verification has traditionally relied on slow, manual, and paper-based processes where employers, embassies, and foreign institutions must wait days or weeks for institutional confirmation. This conventional approach leaves academic credentials vulnerable to physical tampering, digital PDF manipulation, and diploma forgery, creating significant administrative bottlenecks and undermining the global credibility of national degrees.

To address this challenge, **CertiChain** provides a modernized, blockchain-driven solution specifically designed for Sylhet Engineering College (SEC) under the affiliation of SUST. By replacing manual inquiries with cryptographic commitments on the Polygon Amoy blockchain and decentralized IPFS storage, CertiChain enables instant, sub-second certificate authentication via QR codes while driving down operational costs by over 95%. Furthermore, by incorporating client-side biometric verification and zk-SNARK zero-knowledge proofs, the platform directly solves credential sharing and privacy concerns, establishing an immutable, fraud-proof digital credential infrastructure aligned with Bangladesh's national vision for secure digital transformation.

4.6 Justification of this Project

The development of CertiChain is primarily justified by the urgent need to address the severe financial, security, and privacy limitations inherent in current academic credentialing systems.

The project is motivated by the following critical factors:

- **Eliminating High Operational Costs and Inefficiencies:** Traditional paper-based credential management places a massive economic burden on educational institutions. Over the past 15 years, these systems have incurred costs exceeding 2,559 million Taka (approximately 170.5 million Taka annually) due to expenses related to paper, secure printing, postage, and manual verification labor. CertiChain justifies its implementation by reducing these operational expenses by 95.4%, utilizing the highly efficient Polygon Layer-2 network and decentralized IPFS storage to bring annual costs down to approximately 7.9 million Taka.
- **Combating Credential Fraud:** Academic forgery is a global crisis, with approximately 30-40% of job applications containing some form of credential fraud. Conventional physical certificates and static digital PDFs are highly vulnerable to unauthorized modification using common editing software. This project provides a robust solution by mathematically anchoring credentials to the Polygon Amoy blockchain using Keccak-256 cryptographic hashes, making undetected tampering impossible.
- **Solving the Identity Impersonation Gap:** Existing blockchain systems can successfully prove a certificate is authentic, but they fail to verify if the individual presenting the document is the legitimate owner. CertiChain bridges this critical gap through the integration of a hybrid edge-server ArcFace biometric architecture and active liveness detection, directly preventing credential sharing and proxy verification.
- **Protecting Student Privacy:** Storing complete student academic records directly on a public blockchain exposes sensitive personal information to all network participants. This project justifies its advanced architectural design by incorporating Groth16 zk-SNARK zero-knowledge proofs. This allows graduates to mathematically prove specific claims—such as meeting a CGPA threshold or confirming their graduation year—

without unnecessarily revealing their complete academic history to third-party verifiers.

- **Providing Controlled Institutional Governance:** Unlike standard decentralized platforms that lack administrative oversight, this project acknowledges the reality of human error in university administration. CertiChain introduces a multi-tier Role-Based Access Control (RBAC) governance workflow, allowing academic controllers to securely revoke, burn, and re-mint faulty certificates while maintaining a transparent on-chain audit trail.

4.7 Research Gap

The literature suggest that existing systems frequently focus on one or two aspects of credential security rather than the entire lifecycle. A system may provide blockchain verification but lack privacy controls; another may provide privacy-preserving proofs but not institutional correction governance; another may use IPFS but place too much information on-chain.

- Gap 1 — Lifecycle governance: Many credential prototypes do not describe a detailed student → departmental reviewer → central administrator correction workflow.
- Gap 2 — Privacy and verification together: Full certificate disclosure is often used even when only a small claim needs to be verified.
- Gap 3 — Gas efficiency: Storing full academic data on-chain is unnecessary for many verification scenarios.
- Gap 4 — Practical identity checking: Cryptographic validity alone does not prove that the person presenting the credential is the person named on it.
- Gap 5 — Institutional usability: A technically secure system may still fail if students, faculty, and employers cannot use it without understanding blockchain internals.

4.8 Positioning of CertiChain

CertiChain positions itself as an integrated institutional platform rather than a single-purpose blockchain contract. The contribution is the architecture that connects compact blockchain commitments, IPFS, a relational operational database, role-based governance, privacy-preserving proofs, and human-friendly verification interfaces into one lifecycle.

Dimension	Traditional System	Basic Blockchain System	CertiChain
Integrity	Central database / paper	Blockchain record	Blockchain hash commitment + IPFS/database consistency
Verification	Manual institutional contact	QR/token/hash	QR, serial, registration, hash, zk-proof, optional face check
Privacy	Depends on institution	Often limited	Selective zero-knowledge claims
Storage	Centralized files	Often on-chain or centralized	Hybrid: Polygon + IPFS + SQLite
Governance	Administrative office	Usually contract owner	Student → Minter → Admin
Correction	Overwrite or reissue	Varies	Revocation/burn + corrected re-mint
Gas strategy	Not applicable	Can be high	32-byte commitments + bulk Merkle strategy

4.9 Contribution to the Bangladeshi Education Sector

The proposed CertiChain project has the potential to significantly impact the Bangladeshi education sector by:

- **Modernizing Certificate Management:** Introducing advanced technology to streamline and secure certificate issuance and verification processes.
- **Supporting Digital Transformation:** Aligning with national initiatives toward digitization and contributing to the development of digital infrastructure in education.
- **Providing a Scalable Model:** Establishing a framework that can be expanded to other applications within the education system or replicated in similar contexts in other developing countries.

Chapter 5

Requirement Analysis

This chapter defines the functional and non-functional requirements of CertiChain. The requirements are derived from the current project summary and refined using the workflow structure of the supplied pre-defense report. The platform contains four principal interfaces, each with a clear responsibility.

5.1 Stakeholders and Actors

Actor	Primary Responsibility	Main Access
Public Verifier	Validate a credential or a specific qualification claim	Public verification portal
Student	Access credentials, results, transcript, proofs, and submit requests	Student portal
Department Minter	Prepare and endorse institutional academic data	Minter terminal
Academic Admin Controller	Approve governance actions and execute sensitive on-chain operations	Admin portal
Signatory Officer	Review documents in queue, endorse/digitally sign credentials, and manage digital signature/identity	Signatory portal

5.2 Software Requirements Specification (SRS)

The system operates through distinct interfaces serving university administrators (Admins and Department Minters), students, and verification authorities.

5.1.1 Functional Requirements

5.1.1.1 Administrative Interface

The Administrative Interface is the central governance module of CertiChain, designed for university administrators to manage academic credentials, student records, staff privileges and institutional activities. Administrators authenticate through Web3 wallets such as MetaMask, while operational data is maintained off-chain and credential integrity is secured through Keccak-256 hashes stored on the Polygon Amoy blockchain.

The Admin Portal mainly consists of the following modules:

- **Dashboard:** Displays issued certificates, registered students, faculty/staff activities, pending revocations and recent activities.
- **Action Center:** Supports degree minting, staff privilege management, cohort results, semester grades, profile/grade differences, download passes and certificate revocation.
- **Student Records:** Provides access to student academic records, registration information, CGPA, credits, sessions and dossiers.
- **Department Staff:** Manages departments, staff roles and status, and authorization of new staff.
- **Notices:** Allows administrators to create and manage institutional notices with optional attachments.
- **Audit Trails:** Records system events, smart-contract activities, student actions, authentication sessions, governance, RBAC and student-record activities.
- **Settings:** Manages institutional logo and official signatures, including Principal, Examination Controller and Department Heads signatures.
- **My Profile:** Displays administrator information, wallet address, signature and recent activities.

5.1.1.2 Student Interface

The **Student Interface** provides students with secure access to their academic credentials and related services. Students register using their email, ID and password and verify their account through a 6-digit email OTP. After authentication, the Student Portal provides access to blockchain-anchored certificates and academic information associated with the student's wallet.

The main modules of the Student Portal are:

- **Dashboard:** Displays profile information, registration, session, email, phone, DOB, parents' names, institution, CGPA, total credits, degree status and important notices.
- **Certificate:** Allows students to view, download, print and share certificates.
- **Grade Sheet:** Provides grade-sheet viewing, downloading, sharing and official grade-sheet clearance requests with a reason.
- **Document Vault:** Provides secure access to academic documents.
- **Request Center:** Supports academic record correction, duplicate certificate reissue, MOI, sealed envelope services and viewing request decisions.
- **Profile:** Contains academic records and personal/contact information.
- **Privacy & Sharing:** Allows students to generate zk-SNARK proofs for selective disclosure, such as proving a CGPA requirement without revealing unnecessary academic information.
- **Revocation Request:** Allows students to submit up to two lifetime revocation requests when an error is identified.

5.1.1.3 Verification Interface

The Verification Interface allows external users to verify academic certificates without registration or wallet authentication. Verifiers can check a credential using a certificate hash/code, QR code or uploaded verification file. The system validates the credential against its blockchain record and returns its authenticity and current status. This provides a simple and fast way to independently verify academic credentials.

The Verification Interface enables fast, registration-free certificate validation through blockchain records, with QR scanning, certificate hash/code entry or verification-file upload providing convenient access methods for external verifiers.

5.1.1.4 Signatory Interface

The **Signatory Portal** is designed to help authorized officers review documents, manage their digital signatures and complete the endorsement process securely. The interface is divided into three main sections:

- **Review Queue:** Displays documents that are **awaiting the officer's signature** and provides access to the **endorsement history** for previously reviewed or signed documents.
- **My Signature:** Allows the officer to **upload or manage their digital signature** and maintain their **officer identity** information.
- **Overall Purpose:** The portal provides a centralized interface for signatories to verify their identity, review pending documents, apply signatures and track their endorsement activities.

5.1.1.5 Minter Interface

The **Minter Portal** is a centralized interface for authorized institutional minters to manage student information, academic records, certificates, grade sheets and blockchain-based issuance activities.

- **Dashboard:** Provides an overview of registered students, issued degrees, pending grades, pending batches and recent activities.
- **Student Ingestion:** Supports student data import through **CSV/XLSX**, manual entry and archiving of **legacy graduates**.

- **Student Records:** Displays session-based student information including name, registration number, academic session, total credits, CGPA, portal status and academic actions such as **Edit Record** and **Dossier**.
- **Certificate & Ledger:** Manages batch certificate generation, single-student certificate auditing, provisional certificate assignment/generation, verification, issued certificate

ledger and batch candidate verification/review.

- **Grade Sheets:** Supports batch-based semester grade issuance and single-student grade-sheet creation with course and credit information, along with a published ledger.
- **Student Requests:** Centralizes **all, pending, approved and rejected** requests, including profile updates, grade changes, provisional certificates, revocations, leaves and clearance requests.
- **Storage Status:** Provides information about the system's document/data storage status.
- **My Profile:** Allows the minter to manage personal and institutional information, including photo, name, department, email, wallet address, institutional privileges and digital signature through **PNG upload or drawing**.
- **Institutional Privileges:** Defines authorized capabilities such as provisional certificate minting, student batch ingestion, semester grade-sheet management, cross-department signing, emergency revocation, monthly minting quota extension and institutional audit inspection

5.1.2 Academic Results Management:

Full 8-semester course ledgers, credits, and SGPA tracking were not explicitly detailed as a distinct requirement.

5.1.3 Official Transcript Generation:

Standalone institutional transcript generation with grading scales and authority signatures was missing.

5.1.4 Target-Scoped Notifications:

System-wide, role-based real-time alerts (Student, Minter, Admin) were omitted.

5.1.5 Download Quota & Pass Controls:

The institutional download quota mechanism (e.g., 3 free downloads, provisional timers, and administrative quota extension requests) was missing.

5.1.6 Formal Audit Logging:

Explicit functional tracking of security, smart contract, and institutional database events was not separated.

5.1.7 Granular Separation of Workflows:

ZK-Proof generation vs. ZK-Proof verification, as well as Minter endorsement vs. Central Admin approval/burn/re-mint actions, were merged rather than defined as independent functional components.

5.2 Non-Functional Requirements

The non-functional requirements of the platform define the critical quality attributes necessary to ensure a robust, enterprise-grade academic credentialing ecosystem. These requirements are categorized into key operational domains:

Security and Privacy: The system establishes stringent security protocols by utilizing Web3 wallet authentication, rigid role-based permissions, and controlled smart contract access to prevent unauthorized operations. To ensure maximum data privacy, the architecture strictly avoids the unnecessary public exposure of sensitive academic records, instead leveraging zero-knowledge proofs (zk-SNARKs) that allow users to mathematically verify specific qualifications without revealing underlying personal data.

Performance and Scalability: To seamlessly integrate into practical employment and university admission workflows, the system is optimized for high performance, ensuring that common credential verification operations are executed and resolved within seconds.

Furthermore, the platform is highly scalable; by combining compact on-chain cryptographic commitments with off-chain file storage, the system avoids blockchain bloat and minimizes transaction overhead.

Usability and Availability: The platform prioritizes a frictionless user experience, intentionally abstracting complex Web3 mechanics so that students, administrators, and third-party verifiers can complete standard workflows without requiring any detailed blockchain knowledge. System availability is guaranteed by utilizing decentralized, content-addressed storage mechanisms (such as IPFS), ensuring that credential files and their corresponding cryptographic integrity anchors remain permanently retrievable regardless of central server status.

Maintainability and Auditability: For long-term software sustainability, the system enforces high maintainability through a modular Next.js architecture, utilizing strongly typed APIs, Prisma database models, and isolated Web3 services for clean code separation. Finally, the system ensures strict institutional auditability, guaranteeing that all critical lifecycle events—including credential issuance, profile corrections, revocations, and governance approvals—are permanently and transparently traceable.

5.3 Activity Diagram

5.3.1 Student Registration Workflow

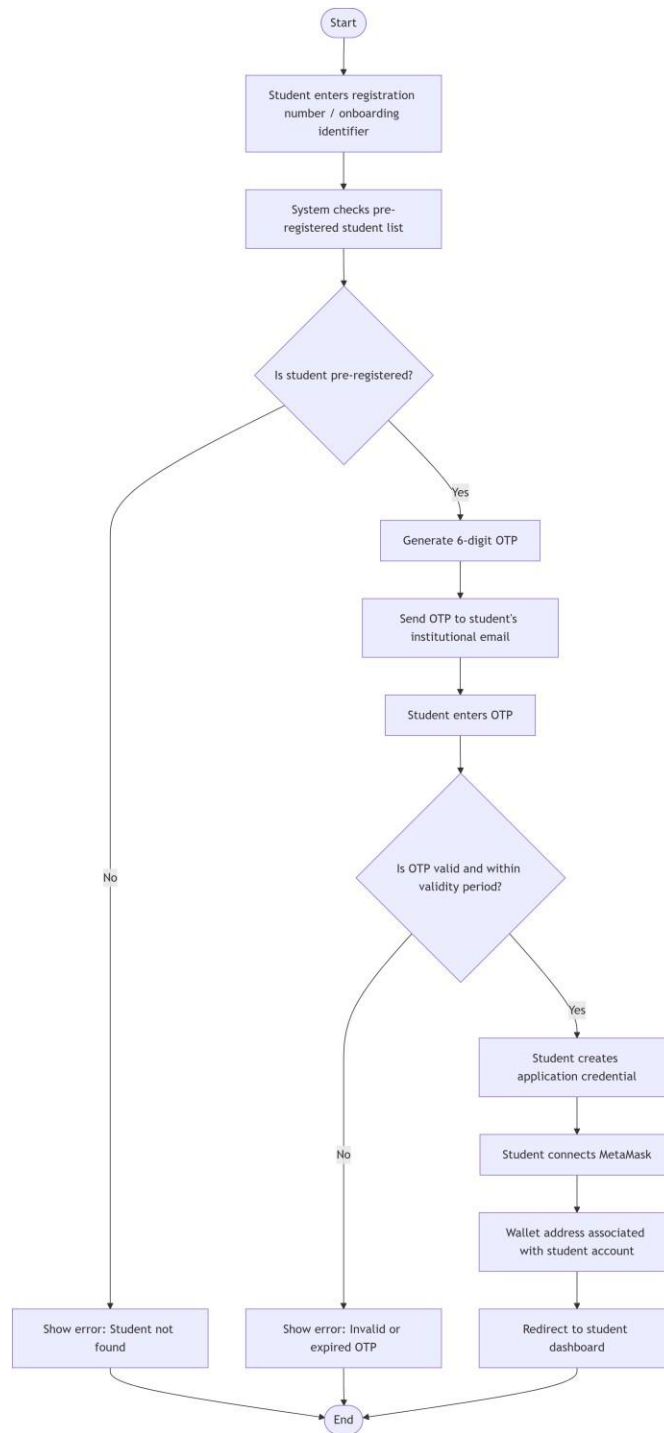


Figure: 5.1 Student Registration Workflow

1. The student enters the registration number or required onboarding identifier.
2. The system checks the pre-registered student list.
3. A six-digit OTP is sent to the student's institutional email.
4. The student enters the OTP within the permitted validity period.
5. The student creates the application credential and connects MetaMask.
6. The wallet address is associated with the student account.
7. The student is redirected to the student dashboard.

5.5 Certificate Issuance Workflow

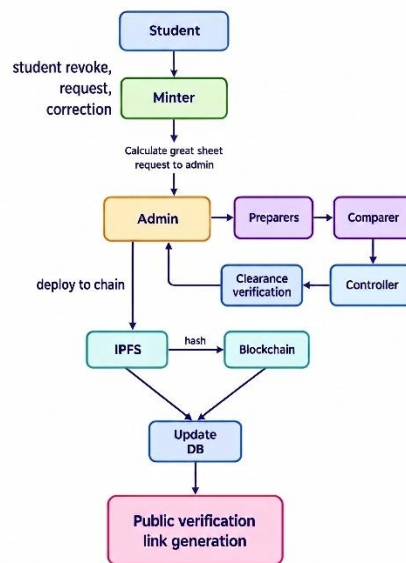


Figure: 5.2 Certificate Issuance Workflow

1. Student academic information is loaded from the institutional/pre-registration record.
2. The authorized department minter reviews the relevant academic data.
3. Certificate metadata is standardized and a deterministic Keccak-256 hash is generated.
4. The certificate PDF, photo, and metadata are uploaded to IPFS.
5. The compact certificate hash and lifecycle data are written to the CertificateRegistry contract.
6. The application stores the transaction hash and IPFS CID in the operational database.
7. The student can access the issued credential and its verification QR code.

5.6 Public Verification Workflow

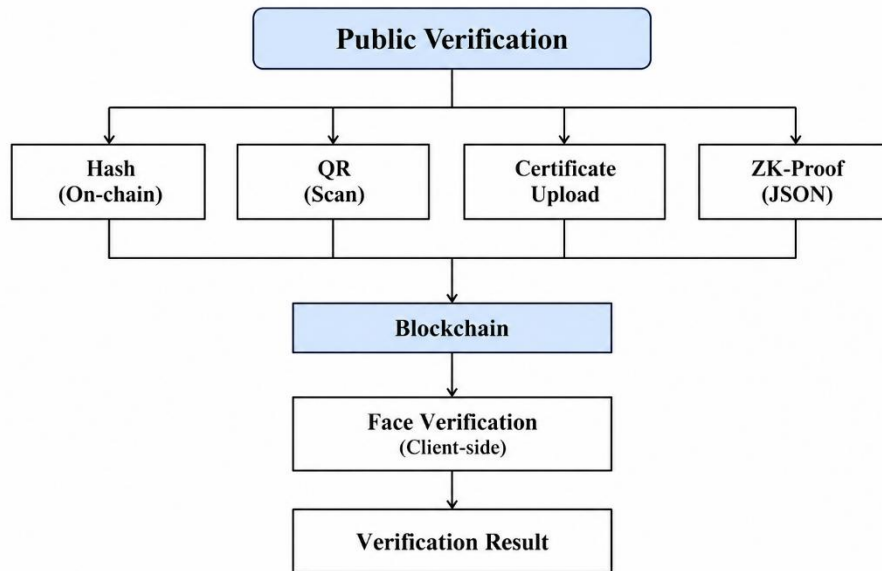


Figure : 5.3 Public Verification Workflow

1. A verifier opens the public verification portal.
2. The verifier provides a serial number, registration number, certificate hash, QR code, or camera input.
3. The application resolves the certificate commitment and queries the blockchain.
4. The system checks existence, issuer authorization, issue time, and revocation status.
5. The application retrieves relevant IPFS content when a document preview is needed.
6. Optional client-side face comparison can be performed using a live webcam image.
7. For a zk-proof submission, the verifier checks the proof and receives only the requested claim result.
8. The portal displays a clear VALID/INVALID result with appropriate non-sensitive details.

5.6.1 University Administration Workflow This models the flow of the Admin and Minter authenticating via MetaMask, processing student data, uploading to IPFS, executing smart contracts on Polygon, and approving revocation requests.

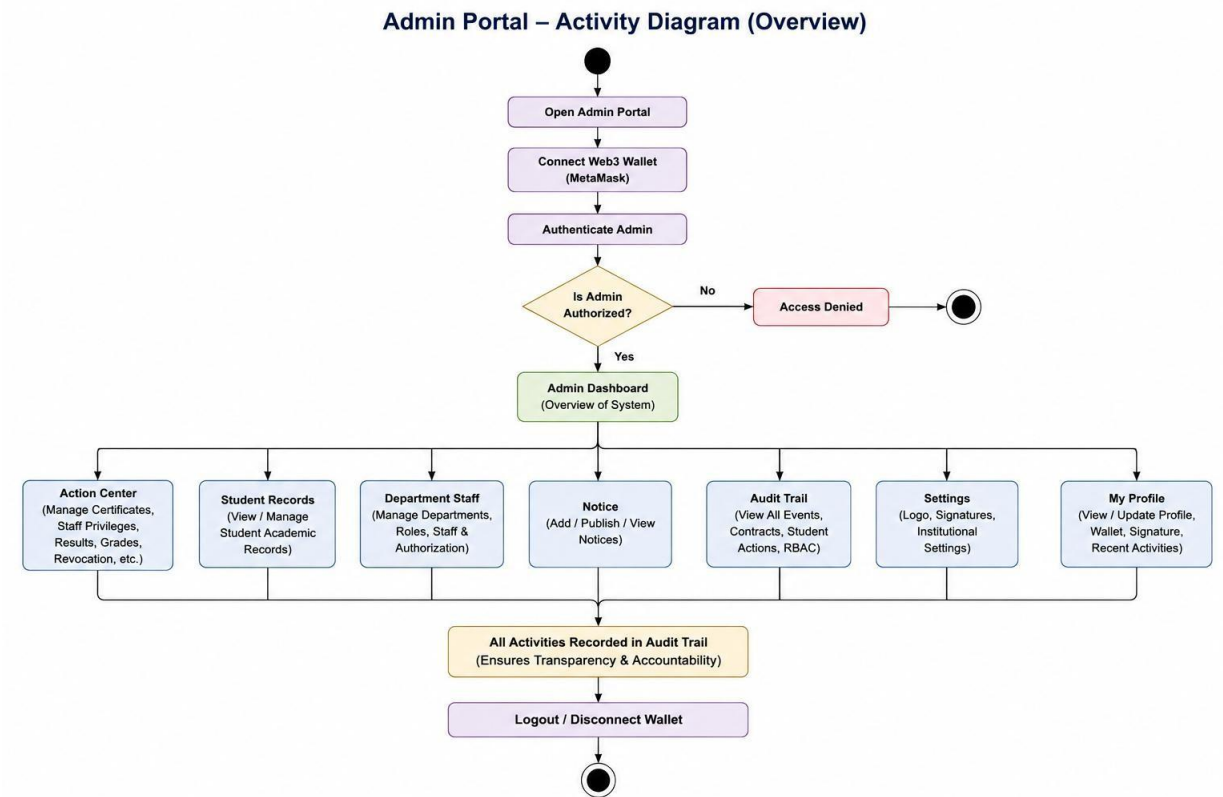


Figure : 5.4 Admin portal activity diagram

5.6.2 Student Interaction Model This models the student completing 2FA registration, connecting their wallet, viewing certificates, downloading PDFs, generating zk-SNARK proofs, and submitting credential correction requests.

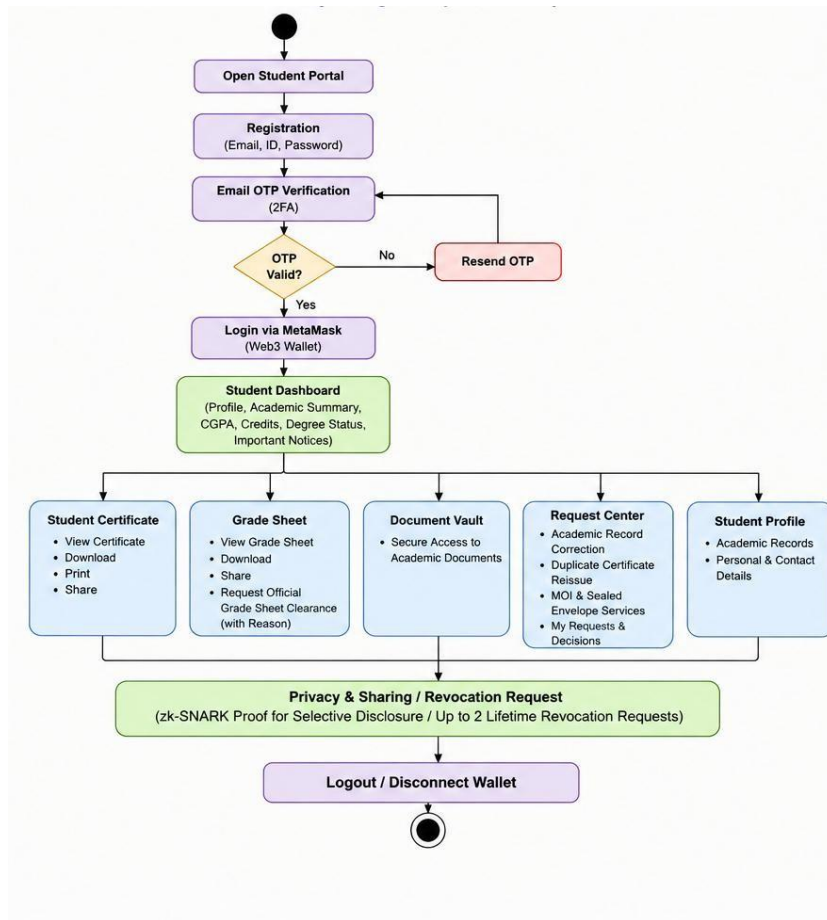


Figure : 5.5 Student portal activity diagram

5.6.3 Verification Authority Processes This models the external verifier accessing the portal, scanning a QR code or uploading a zk-Proof, completing the active liveness biometric challenge, and receiving real-time blockchain validation results.

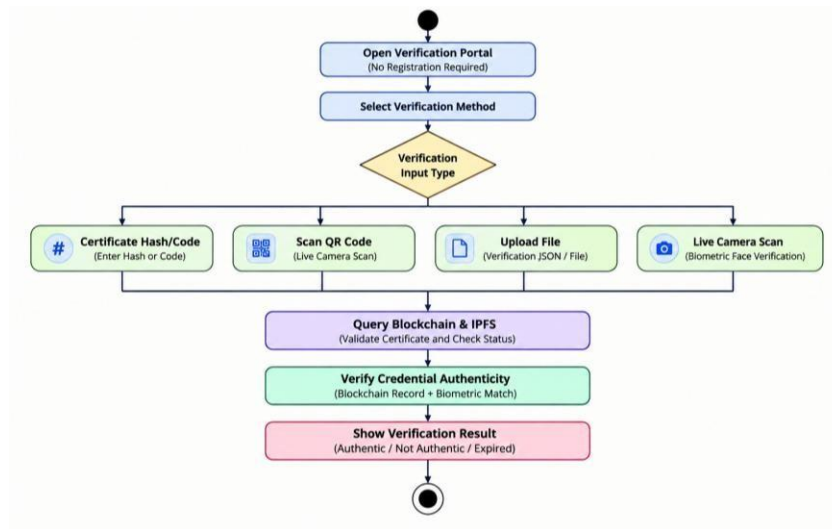


Figure :5.6 Verification portal activity diagram

5.7 Use-Case Model

The system consists of five principal actors: Student, Department Minter, Signatory Officer, Admin Controller, and Public Verifier. Each actor interacts with the system through a dedicated portal according to their assigned role and permissions.

1. **Student Portal :** Register and log in; verify through OTP; manage profile; view and download certificates; view and download grade sheets; access document vault; submit academic record correction, duplicate certificate, MOI, sealed envelope, clearance, and other requests; view request decisions; receive important notices.
2. **Minter Portal:** Register and ingest students through CSV/XLSX/manual entry; manage student records; manage certificate generation and ledger; prepare and issue grade sheets; review and manage student requests; manage legacy graduate archives; view storage status; manage minter profile, privileges, and digital signature.
3. **Admin Portal:** Manage dashboard and system activities; approve degree/certificate issuance; manage staff and institutional privileges; manage student records; approve grades and profile corrections; manage revocations; publish notices; manage system branding and signatures; review audit trail and ledger; manage system settings and admin profile.
4. **Public Verifier Portal:** Verify certificates using certificate hash/code; scan QR codes; upload certificate files for verification; check certificate authenticity and status; retrieve

verification information; perform optional advanced verification such as face comparison or ZK-proof verification.

5. **Signatory Portal:** Access review queue; view documents awaiting signature; review documents; endorse/sign documents; manage digital signature and officer identity; view endorsement history.

5.8 Use-Case Modeling

Use-case modeling analyzes system functionality from the user's perspective, identifying key interactions between actors and the system.

5.8.1 Primary Use Cases The system supports three primary use cases corresponding to the main actors: university administration (Minters/Admins), students, and verif

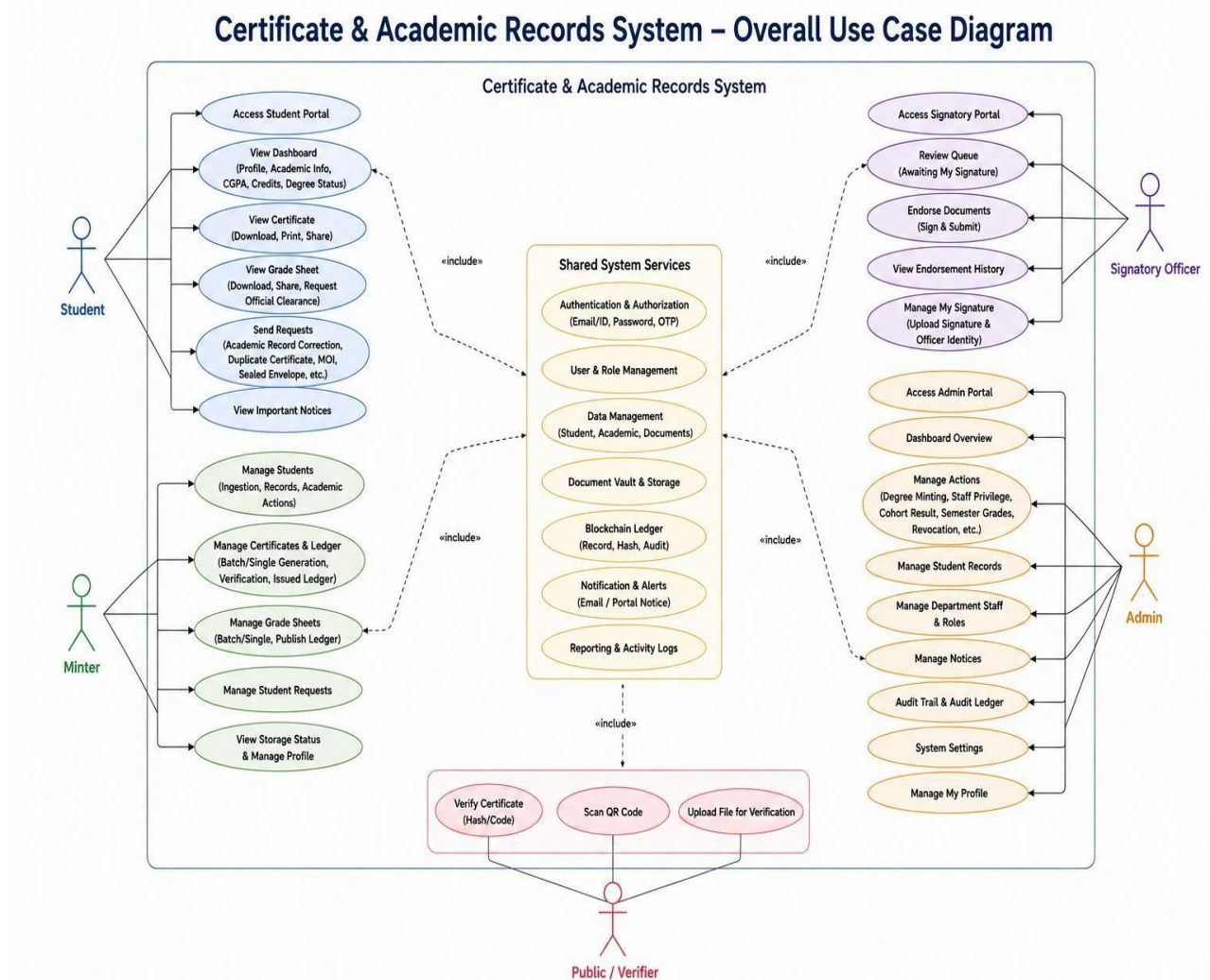


Figure : 5.7 Use Case diagram

5.8.1.1 Use Case 1: Certificate Issuance

- **Actor:** University Administrator / Department Minter
- **Description:** Complete workflow for issuing blockchain-based academic certificates through administrative interfaces utilizing a hybrid storage model.
- **Main Flow:** The administrator authenticates through MetaMask. They input comprehensive student information into the system or upload bulk CSV files for cohorts. The system validates data integrity, saves operational data to the Prisma/SQLite database, and pins certificate PDFs and student photos to IPFS. A cryptographic hash is generated using the Keccak-256 algorithm. Smart contract execution stores this 32-byte hash securely on the Polygon Amoy blockchain, linking the unique identifier to the student's designated wallet address while ensuring ultra-low gas fees.

5.8.1.2 Use Case 2: Certificate Access and Management

- **Actor:** Student
- **Description:** Student interactions for accessing, viewing, revoking, and privately sharing academic credentials.
- **Main Flow:** The student initiates wallet connection through MetaMask. The system queries the Polygon blockchain and fetches off-chain metadata from IPFS/SQLite to display certificate details on the dashboard. The student selects a desired credential for viewing, downloading as an embedded QR-PDF, or sharing. If privacy is required, the student activates the zk-SNARK engine to generate a zero-knowledge proof payload (e.g., proving graduation without revealing grades). The student may also initiate a credential revocation workflow if they detect typographical errors.

5.8.1.3 Use Case 3: Certificate Verification

- **Actor:** Verification Authority (Employer/Institution/Embassy)
- **Description:** Verification process enabling third parties to authenticate both certificate legitimacy and physical identity through blockchain and biometric validation.

- **Main Flow:** The verifier accesses the public portal and inputs a token, scans a QR code, or uploads a zk-Proof JSON. The system executes a smart contract read-query to retrieve the credential's active status from Polygon. Concurrently, the system prompts a biometric liveness challenge (face centering, turning, blinking) via the verifier's webcam. A client-side ArcFace neural network compares the live facial embedding against the IPFS-stored student photo. Verification results are displayed in real-time, indicating both the cryptographic validity of the certificate and the biometric match confidence score.

5.9 Security Requirements

- Only authorized issuer wallets can invoke certificate issuance functions.
- Administrative functions must be protected by owner-level authorization.
- Certificate hashes must be deterministic so that the same standardized data produces the same commitment.
- Revoked certificates must not be accepted as valid.
- Student corrections must not silently replace historical credential state.
- Sensitive student data must not be unnecessarily exposed on the public verification portal.
- Biometric processing should remain client-side according to the project design.
- Private wallet keys must never be requested, stored, or transmitted by the application.

Chapter 6

System Design

The system design follows the hybrid architecture described in the current project summary. The design separates the application layer, database layer, IPFS layer, blockchain layer, and browser-side privacy/biometric functions. This separation allows the platform to use each technology for the problem it solves best.

5.1 System Architecture

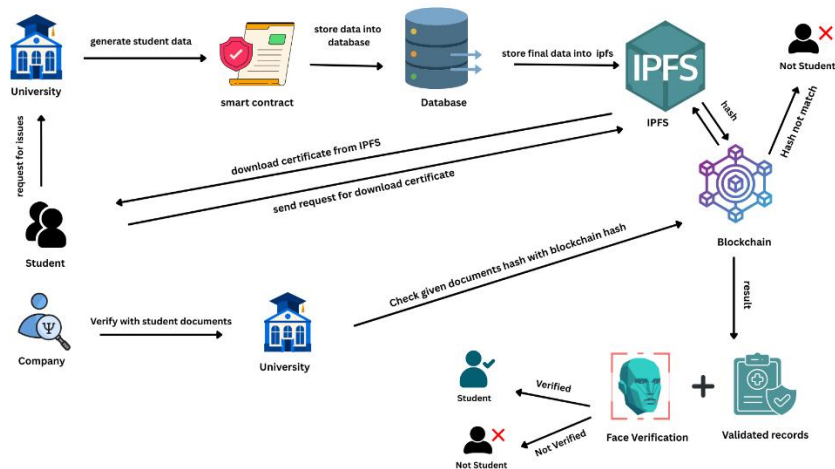


Figure 6.1: System Architecture

6.1.2 Architecture Components

1. **Student Layer:** View and download certificates. Share verifiable proofs while hiding specific grades/GPA using ZK-SNARKs. Perform client-side facial comparison for identity verification.
2. **Verifier Layer:** Allow third parties (employers/companies) to verify certificate authenticity. Match on-chain hashes and check revocation/expiry status.
3. **Administrative Layer:** University admin panel. Manage student records, grades, handle certificate issuance requests, and store audit trails/logs.
4. **Signatory Layer :** The Dean/Registrar approval layer. This is where authorized personnel review and digitally approve a certificate before it is minted. May include multi-signature workflows.

5. **Minter Layer:** The backend issuance service. Automatically triggers after approval. It uploads the certificate PDF to IPFS, constructs the metadata, manages gas fees, and submits the transaction to Polygon Amoy.
6. **Blockchain Layer:** The core immutable ledger. Stores certificate hashes, issuance timestamps, issuer addresses, and revocation states on-chain. IPFS acts as the decentralized file storage companion.

6.2 Data Flow Diagram

The primary data flow begins with student and institutional data entering the application. Academic information is stored in the operational database, while credential files are uploaded to IPFS. The application then creates a deterministic certificate hash and sends the compact commitment to the Polygon smart contract. During verification, the process is reversed: the verifier supplies an identifier, the application resolves the credential, the blockchain confirms the integrity and status, and IPFS provides the document or image when required.

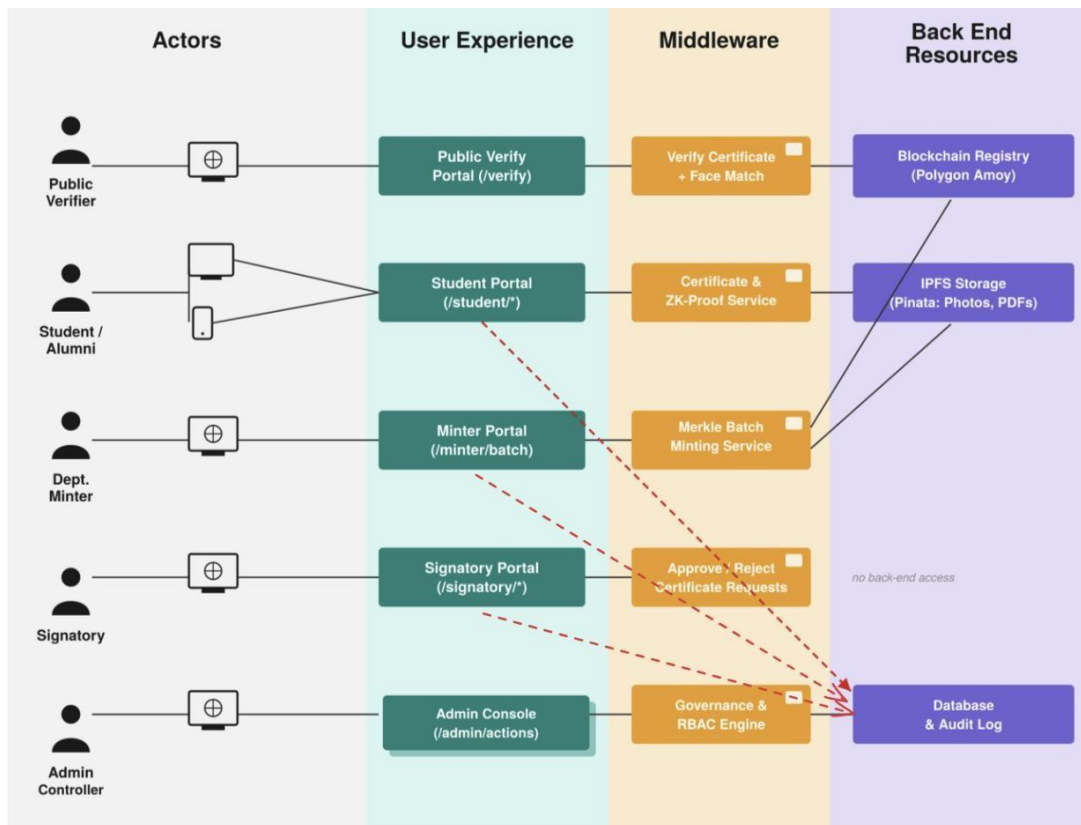


Figure 6.2: Certificate issuance and verification data flow.

6.3 On-Chain Data Design

The CertificateRegistry contract is deliberately small. The core record contains only the fields needed for integrity and lifecycle verification.

```
struct CertificateRecord {
    bytes32 certHash;
    address issuer;
    uint256 issuedAt;
    bool isRevoked;
    uint256 revocationCount;
}

mapping(address => bool) public authorizedIssuers;
mapping(bytes32 => CertificateRecord) public certificates;
```

This design prevents the smart contract from becoming a large academic database. It also makes verification logic simple: the verifier can ask whether a certificate hash exists, which address issued it, when it was issued, and whether it has been revoked.

6.4 Smart Contract Functions

Function	Access	Purpose
issueCertificate	Authorized issuer / owner	Create a new certificate commitment if the hash is not already registered.
revokeCertificate	Authorized issuer / owner	Mark an existing certificate as revoked and increment revocation count.
verifyCertificate	Public view	Return existence, issuer, issue time, and revocation status.
authorizeIssuer	Owner	Grant certificate issuance permission to an institutional wallet.
Issuer management	Owner	Maintain the trusted issuer set and preserve governance control.

6.5 Gas Minimization Strategy

Gas optimization is a central design goal. The project summary specifies four main strategies. First, only a 32-byte certificate commitment is stored on-chain. Second, large files remain in IPFS. Third, client-side hashing avoids unnecessary string manipulation inside the smart contract. Fourth, Merkle-tree commitments can support bulk cohort registration.

Strategy	Why it helps
32-byte bytes32 commitment	Minimizes persistent storage compared with storing long strings or full JSON.
IPFS for heavy files	Avoids blockchain storage of PDFs, photos, and metadata.
Client-side hashing	Moves deterministic hashing work out of contract execution.
Merkle batch commitment	Allows many student records to be represented by one root commitment.
Compact contract state	Keeps read/write operations focused on certificate integrity and lifecycle.

6.6 Database Architecture

The current project specification uses Prisma ORM 7.x with SQLite. The database stores application information that does not need blockchain-level immutability, including student accounts, pre-registration data, course ledgers, requests, notifications, and audit information.

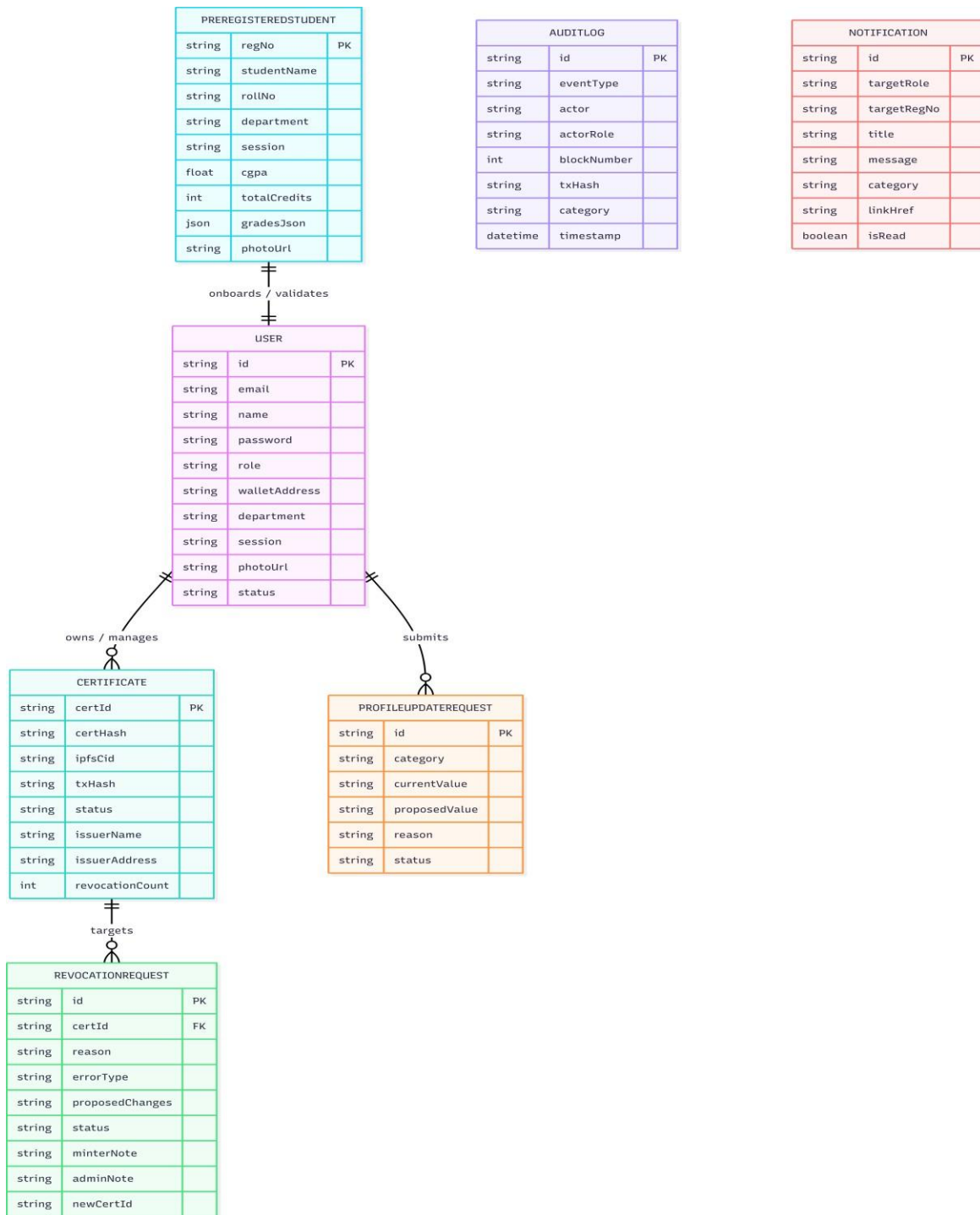


Figure: 6.3 Database entity relationship overview.

6.7 High-Level Design

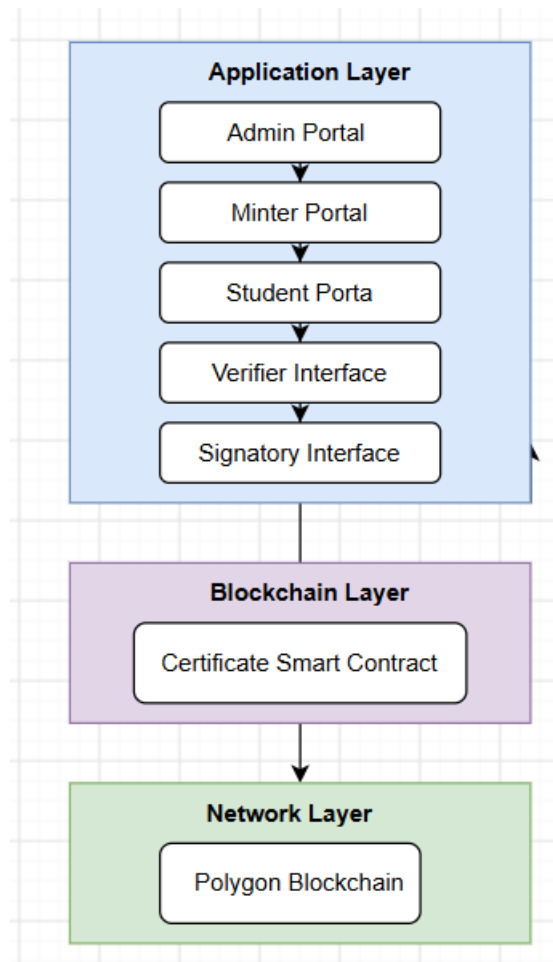


Figure 5. High-Level Design

At the highest level, CertiChain can be viewed as five cooperating subsystems: the user-facing application, the operational database, decentralized file storage, the blockchain registry, and privacy/identity verification modules. The application layer coordinates these components and enforces institutional business rules.

6.8 Low-Level Design

6.8.1 Frontend Component Design

The frontend is implemented with Next.js using an App Router architecture. Separate route groups are used for public verification, student functions, departmental minting, and central administration. Reusable components handle forms, tables, status badges, certificate previews, QR display, modal dialogs, notifications, and blockchain transaction states.

6.8.2 Web3 Integration

wagmi and viem provide the Web3 integration layer. MetaMask is used as the wallet connector and transaction signer. The application never needs the student's private key. Contract reads can be performed without a transaction, while state-changing operations require wallet confirmation.

6.8.3 IPFS Integration

IPFS integration is isolated behind application services so that the rest of the system can work with file references rather than provider-specific upload logic. This improves maintainability and makes it possible to change pinning providers in the future without redesigning the credential model.

6.8.4 Security Design

- Authentication: application credentials plus wallet binding for students and Web3 actors.
- Authorization: role and wallet checks at the application and contract levels.
- Integrity: deterministic Keccak-256 commitments.
- Storage security: large files separated from the on-chain state.
- Privacy: selective zk-proof claims and limited public result display.
- Identity assurance: optional client-side face comparison.
- Governance: multi-step review before sensitive credential lifecycle changes.

6.9 Certificate Lifecycle

A certificate is not treated as an unchangeable file with no lifecycle. The system explicitly models the possibility that a credential may later be found to contain an error or become compromised. The lifecycle therefore includes active issuance, verification, revocation, burn/replacement, and historical audit events.

Chapter 7

Methodology and Development Planning

The development methodology combines requirements analysis, iterative system design, smart contract development, frontend implementation, integration, testing, and documentation. Because blockchain transactions are difficult to modify after deployment, the methodology places additional emphasis on early validation of contract logic and data structures.

7.1 Research Methodology

9. Problem identification: analyze weaknesses of traditional certificate issuance and verification.
10. Requirement extraction: identify stakeholders, security needs, privacy requirements, and institutional workflows.
11. Architecture design: divide data and responsibilities between database, IPFS, blockchain, and browser-side modules.
12. Cryptographic design: define the deterministic certificate hash and privacy-preserving verification claims.
13. Smart contract implementation: develop and test CertificateRegistry.sol.
14. Application implementation: build student, minter, admin, and public verification interfaces.
15. Integration: connect the application to Polygon, IPFS, database, wallet, zk-proof, and biometric modules.
16. Testing: perform unit, integration, workflow, security-oriented, and usability testing.
17. Evaluation and documentation: analyze the architecture, limitations, contribution, and future expansion.

7.2 Development Lifecycle

The project follows an iterative lifecycle rather than a single linear implementation. Each major component is developed and validated before the next dependency is added. This approach reduces the risk of debugging multiple interacting layers at the same time.

7.3 Development Phases

Phase	Main Activities	Primary Output
Phase 1	Requirement analysis and technology study	SRS, architecture decisions, research questions
Phase 2	Database and application foundation	Prisma schema, project structure, role model
Phase 3	Smart contract development	CertificateRegistry.sol and contract tests
Phase 4	Blockchain and wallet integration	Polygon Amoy connectivity, contract reads/writes
Phase 5	Portal implementation	Student, Minter, Admin, and Verifier interfaces
Phase 6	IPFS and document integration	Certificate/document storage and retrieval
Phase 7	ZK and biometric modules	Proof workflows and client-side face comparison
Phase 8	Testing and refinement	Bug fixes, validation, security checks
Phase 9	Documentation and deployment preparation	Final report, user workflow, deployment configuration

7.4 Resource Requirements

Type	Requirement
Processor	Modern Intel Core i5 / AMD Ryzen 5 class processor or better
Memory	At least 8 GB RAM recommended
Storage	SSD recommended for development tools and project dependencies
Operating System	Windows, Linux, or macOS capable of running Node.js and development tools
Development Tools	VS Code, Node.js, npm, Git, browser with MetaMask
Blockchain Tools	Hardhat, Solidity compiler, Polygon Amoy access
Database	SQLite with Prisma ORM
Storage Service	Pinata/IPFS
Browser APIs	WebGL/webcam support for client-side face verification

7.5 Risk Management

- Smart contract bugs: mitigate through small contract scope, access modifiers, and repeated local/testnet testing.
- Wallet/network errors: provide explicit network and transaction states in the UI.
- IPFS retrieval issues: retain CID references and use a resilient pinning strategy.
- Privacy leakage: keep sensitive information off-chain and use selective proof verification.
- Incorrect corrections: require student evidence and departmental endorsement before central approval.
- Biometric mismatch: present biometric verification as an additional signal rather than the only credential authenticity mechanism.
- Database inconsistency: store transaction hashes and certificate commitments so application state can be cross-checked against the chain.

Chapter 8

Development and Implementation

The implementation brings together the technologies specified in the current project summary: Next.js, TypeScript, wagmi, viem, Solidity, Hardhat, Polygon Amoy, Prisma, SQLite, Pinata/IPFS, TensorFlow.js, and face-api. The goal of the implementation is to keep blockchain interactions explicit and auditable while making normal academic workflows feel like a familiar web application.

8.1 Technology Stack

Technology	Version / Role in Current Specification
Next.js	15.5.22 — application framework and routing
TypeScript	5.7.2 — typed application development
wagmi	2.14.0 — wallet and contract integration
viem	2.22.0 — Ethereum-compatible blockchain interaction
Solidity	0.8.20 — smart contract implementation
Hardhat	2.22.17 — contract development and testing
Polygon Amoy	Chain ID 80002 — target test network
Prisma	7.9.1 — ORM and database access
SQLite	Operational relational database
Pinata/IPFS	Decentralized/content-addressed file storage
TensorFlow.js	Browser-side ML runtime for biometrics
@vladmandic/face-api	Face detection and recognition support
Lucide React	Interface icon library

8.2 Smart Contract Implementation

The CertificateRegistry contract is intentionally compact. Its main responsibility is to create and verify certificate commitments and control authorized issuers. The contract does not store the full certificate PDF, transcript, or student photo. This decision is central to both privacy and gas efficiency.

```
contract CertificateRegistry {
    address public owner;

    struct CertificateRecord {
        bytes32 certHash;
        address issuer;
        uint256 issuedAt;
        bool isRevoked;
        uint256 revocationCount;
    }

    mapping(address => bool) public authorizedIssuers;
    mapping(bytes32 => CertificateRecord) public certificates;

    function issueCertificate(bytes32 certHash) external onlyIssuer { ... }

    function revokeCertificate(bytes32 certHash) external onlyIssuer { ... }

    function verifyCertificate(bytes32 certHash)
        external view
        returns (bool exists, address issuer,
            uint256 issuedAt, bool isRevoked) { ... }
}
```

8.3 Frontend Development and User Interface Design

CertiChain is a decentralized academic credential platform engineered for Sylhet Engineering College to eliminate certificate forgery and centralized data tampering. It utilizes a hybrid architecture that pairs 32-byte Polygon blockchain commitments with IPFS storage, reducing on-chain gas costs by over 99%. The system provides five distinct Next.js role-based portals to govern the full institutional issuance and verification lifecycle. It integrates browser-edge zk-SNARKs and ArcFace AI facial verification to ensure privacy-preserving selective disclosure and eliminate credential sharing.

8.3.1 Home Page

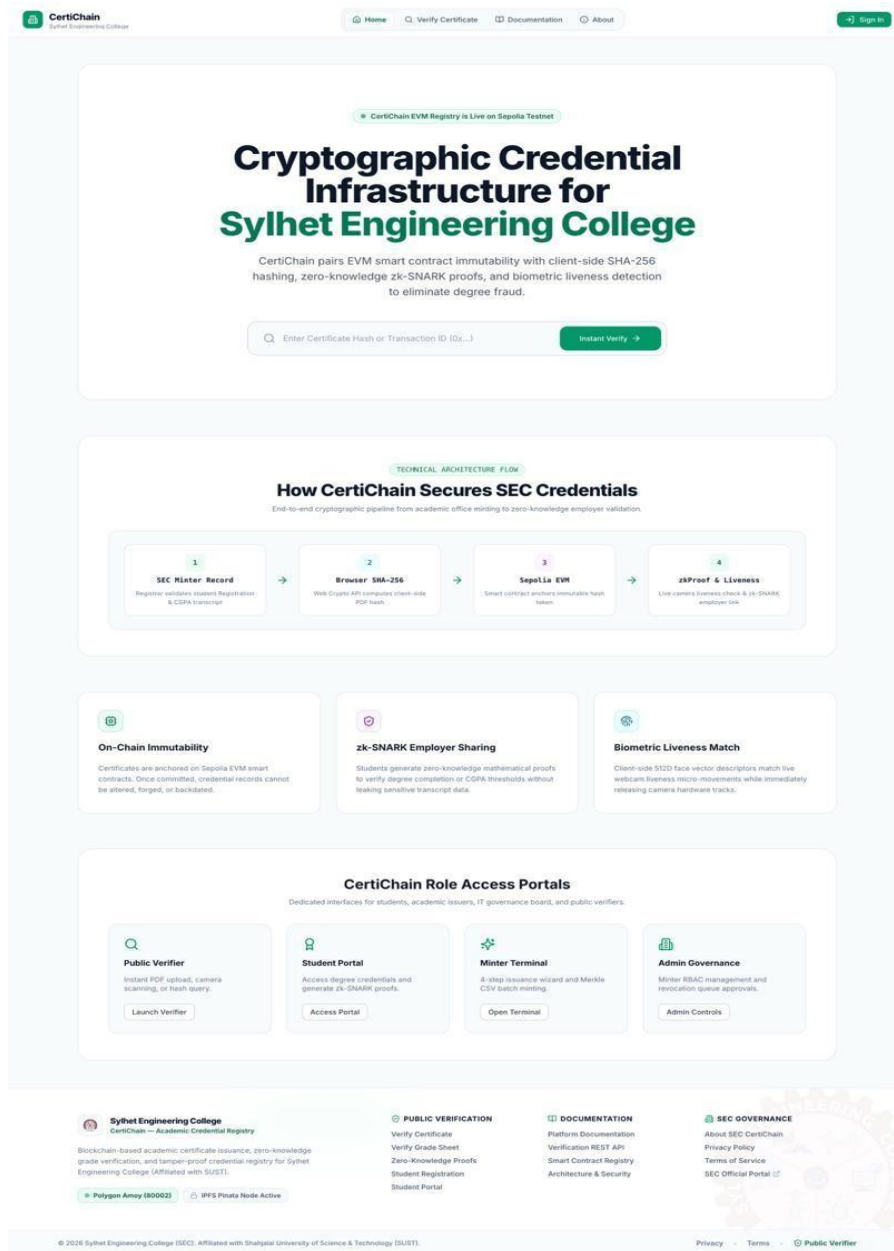


Figure 8.1 Home Page

8.3.2 Public Verification Portal

The public verification portal is deliberately designed so that a verifier does not need a student account or a MetaMask wallet for normal verification. The portal supports several lookup methods because different organizations may have different information available.

- Certificate serial number lookup.
- Registration number lookup.
- Direct certificate hash lookup.
- QR code upload or live camera scan.
- zk-proof JSON verification for selective qualification claims.
- Optional live face comparison against the official student photo.

After a successful lookup, the system checks the blockchain state before presenting the result. A revoked certificate must not be shown as valid. Where appropriate, the portal displays only information needed for verification, reducing unnecessary exposure of personal academic data.

The figure displays two screenshots of the Verifier portal interface.

Left Screenshot: Certificate Verification Result

At the top, there are three tabs: "Certificate Hash / Code", "Scan QR Code", and "Upload File". Below these, a search bar contains the hash "0x31fa184dae3b5c5f050d099591ac168b1154b24ea5f8283622dda770b0ae366". A green "Verify" button is next to the search bar.

The main content area shows a green "Certificate Verified" status with the text "This certificate is authentic and currently active." and an "Active" button.

Below this is a "Provisional Certificate" section for **SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY**, Sythet Engineering College (SEC). It displays the following details:

Student Name	Registration No	CGPA
Omar Faruk	1803010042	3.95 / 4.00

Additional details include:

- Degree: Bachelor of Science in Computer Science & Engineering
- Department: Computer Science & Engineering (CSE)
- Passing Year: 2026 (Session 2022-2023)
- Certificate No: PROV-CSE/2026/0042
- Issued By: Office of the Principal & Controller
- Date Issued: August 20, 2026

At the bottom, there is a "Verification Details" section with the "Verification Code" and a "Verify Identity" button. A "Share Link" and "Print Report" button are also present.

Right Screenshot: Biometric Face Verification

The title is "Biometric Face Verification" with the candidate name "Omar Faruk". There are two tabs: "Live Camera (Recommended)" and "Upload Static Photo".

A yellow notice box states: "Notice: Static photo verification compares mathematical facial embeddings, but does not perform live anti-spoofing checks. Live webcam verification is recommended for highest security."

Below the notice is a dashed box with an upload icon and the text "Click to change verification photo" and "Supported formats: JPG, PNG, WEBP".

There are two photo comparison areas:

- ENROLLED RECORD:** Shows a photo of the student.
- UPLOADED CANDIDATE PHOTO:** Shows a photo of the candidate.

At the bottom, a green box confirms "Facial Identity Match Confirmed" with "Euclidean Distance: 0.7648 (Threshold: 0.50)" and "83.7% Similarity". There are "Cancel" and "Confirm & Accept Verification" buttons.

Figure 8.2 Verifier portal

8.3.3 Student Portal

The student portal is designed as the primary credential management workspace. It provides a certificate viewer, semester grade-sheet center, transcript center, proof-generation area, transfer/MOI functions, correction requests, and revocation claim submission.



Omar Faruk
Computer Science & Engineering (CSE)

REGISTRATION NUMBER (PERMANENT ID)
1803010042

ACADEMIC SESSION
2022-2023

EMAIL
faruk@sec.ac.bd

VERIFIED STUDENT

PHONE
+880 1712-345678

DATE OF BIRTH
2001-05-14

FATHER'S NAME
Md. Nurul Islam

MOTHER'S NAME
Mst. Rezia Begum

INSTITUTION
Sylhet Engineering College

Cumulative CGPA
3.95 / 4.00
Academic Distinction

Completed Credits
160.0
All 8 Semesters Completed

Degree Status
Degree Awarded
B.Sc. in CSE

Certificate
Degree credential

Grade Sheet
Semester SGPA

Document Vault
MOI, Testimonial & Letters

Requests Center
Profile & record claims

Important Notices
Announcements from Sylhet Engineering College administration

2 NOTICES

URGENT

Convocation 2026 Registration Open

All graduating students of session 2022-2023 are requested to register for the upcoming convocation ceremony by August 30, 2026. Registration forms are available at the academic section.

August 20, 2026

Student Revocation:

Report Credential Typo / Discrepancy

Submit an official data correction claim for Academic Council review

TARGET CREDENTIAL
CERT-PROV-1787243602906-1803010042

STUDENT
Omar Faruk (1803010042)

Discrepancy Category *

CGPA / Final Cumulative Grade Point Discrepancy

Current Printed Value (Incorrect)

Proposed Correct Value (Official)

e.g. As shown on certificate

e.g. As in NID / SSC

Detailed Reason & Justification *

Explain the reason for correction and cite official documents (e.g. NID #, SSC Certificate, SUST Admission Record)...

Supporting Proof Document (Optional, PDF / Image)

Attach National ID, SSC or Academic Record

Blockchain Re-Mint Policy:

On-chain tokens cannot be modified in place. An approved request revokes the old token and generates a new cryptographic certificate on Polygon Amoy. Limit: 2 lifetime requests.

Cancel

Submit Discrepancy Report

Endorse Certificate Correction & Revocation

Certificate: CERT-PROV-1787243602906-1803010042

Minter Review

Omar Faruk

Reg: 1803010042

Error Type: CGPA / Grade Calculation Error

Pinpointed Claim: Change "3.50" to "4.00"

Reason:
The main certificate has miss information about my CGPA, which is currently 3.50, but my actual cgpa 4.00.

Attached Proof:
bbauXHfegk...

DEPARTMENTAL VERIFIED VALUES

Corrected Student Name

Corrected Reg No

4.00

1803010042

Corrected CGPA

Passing Year

4.00

2026

Department Verification Note

Verified against CSE departmental records & grade registers.

Reject Request

Cancel

Endorse & Forward to Controller

Certificate: CERT-PROV-1767243662908-1693010042 • Target: Omar Faruk

MINTER ENDORSED

ACTIVE CERTIFICATE ON RECORD

Polygon Amoy

Student Name:

Omar Faruk

Registration No:

1893010042

Department:

Computer Science & Engineering--

CGPA on Record:

3.95

Passing Year:

2026

Kecck-256 Hash:

8x31Fa18A24x3B5c5F95A0989...

STUDENT CLAIM & EVIDENCE

2026-08-29

Error Category:

CGPA / Grade Calculation Error

Claimed Correction:

Change "3.50" to "4.00"

Student Reason / Explanation:

"The main certificate has miss information about my CGPA, which is currently 3.50, but my actual cgpa 4.00."

Supporting Document:

certificate--

DEPARTMENTAL MINTER VERIFICATION

Endorsed • Department Head (Minter) (CSE)

Reviewing Minter: Department Head (Minter) (CSE)

Endorsement Time: 8/21/2026, 2:36:04 AM

Department Note: "Verified against CSE departmental records & grade registers."

CORRECTED CERTIFICATE DATA (TO BE RE-MINTED ON POLYGON AMOY)

Atomic On-Chain Burn & Mint

Corrected Full Name *

4.00

Registration No *

1893010042

Roll No

2022331001

Department

Computer Science and Engineering

CGPA *

4.00

Passing Year *

2026

Admin Governance Note / Justification *

Approved by Academic Controller. Minter endorsement [Department Head (Minter) (CSE)] verified against records.

Decline Request

Cancel

Revoke Only

Execute On-Chain Burn & Re-Mint

Student Certificate Issuance

CertiChain

Sylhet Engineering College

Student Portal

Active

Dept:

Computer Science & En...

STUDENT PORTAL

Dashboard

Certificates

Grade Sheet

Requests

Academic Certificates

Official blockchain-verified degree & provisional academic credentials

Academic Certificate Issuance Pipeline

AWAITING MINTER PROPOSAL

Multi-signatory governance, cryptographic Kecck-256 verification, and Polygon Amoy on-chain commitment.

Refresh Status

OFFICIAL ACADEMIC COUNCIL ISSUANCE STAGES

STAGE 1

Minter Academic Audit

Department proposal submitted

STAGE 2

Institutional Review

Super Admin initial check

STAGE 3

Signatory Committee

Multi-officer signatures

STAGE 4

Polygon Amoy & IPFS

Smart contract deployment

STAGE 5

Conferred & Published

Available for QR verification

CANDIDATE VERIFICATION RECORD

CANDIDATE NAME

Tariqul Islam Sunny

REGISTRATION NO

1893010043

PASSING YEAR

2026

DEPARTMENT

Computer Science & Engineering (CSE)

ACADEMIC SESSION

2022-2023

DEGREE TITLE

Bachelor of Science in Engineering

CUMULATIVE CGPA

3.88

TOTAL CREDITS

168 Credits

Need interim proof of graduation?

Apply for a Provisional Certificate for immediate job joining, scholarship, or foreign university applications.

Apply for Provisional Certificate

Sylhet Engineering College

CertiChain — Academic Credential Registry

Blockchain-based academic certificate issuance, zero-knowledge grade verification, and tamper-proof credential registry for Sylhet Engineering College (Affiliated with SUST).

Polygon Amoy (80002)

IPFS Pinata Node Active

PUBLIC VERIFICATION

Verify Certificate

Verify Grade Sheet

Zero-Knowledge Proofs

Smart Contract Registry

Student Registration

Student Portal

DOCUMENTATION

Platform Documentation

Verification REST API

Smart Contract Registry

Architecture & Security

SEC GOVERNANCE

About SEC CertiChain

Privacy Policy

Terms of Service

SEC Official Portal

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Privacy

Terms

Public Verifier

Figure 8.3 Student portal

8.5 Department Minter Terminal

The Department Minter Terminal is a departmental governance layer. A minter is not given unrestricted administrative control. Instead, the minter manages departmental student data, prepares academic results, checks supporting records, and endorses requests for the central academic controller.

DH

Department Head (Minter) AUTHORIZED MINTER
Computer Science & Engineering (CSE) • Sylhet Engineering College
mojunder@sec.ac.bd

Special Institutional Privileges
Request elevated smart contract permissions and emergency signing authority from Root Administration

Provisional Certificate Minting Authority GRANTED
Permission to issue fast-track interim and provisional degree credentials prior to convocation.
Active Authority

Student Batch Ingestion Access GRANTED
Authority to upload bulk student CSV batches and manage the departmental pre-registration whitelist.
Active Authority

Semester Grade Sheets Management GRANTED
Permission to upload, verify, and modify 8-semester course credits and SGPA transcripts.
Active Authority

Cross-Department Signing Privilege NOT GRANTED
Authority to co-sign inter-disciplinary and dual-degree engineering awards across SEC departments.
Request Privilege

Digital Signature
Official endorsement signature stamped on certificates & grade sheets


Draw / Replace Upload PNG
Transparent PNG or SVG. Used on endorsed grade sheets and departmental credentials.

Departmental Authority
Academic issuance scope & clearance

Minter Certificate Generation

Institutional Certificate Center & Ledger
Cohort batch minting, individual student academic audits, and master on-chain certificate ledger.

Batch Generation

Single Student Audit

Issued Ledger

Batch Certificate Issuance
Select target session and assign Provisional or Main Degree certificates — Computer Science & Engineering (CSE)

Certificate Type

Academic Session

Import Results Spreadsheet

Provisional
Academic completion

Main Degree
Official convocation

2022-2023

Sync DB

Upload CSV / Excel

Name, Registration No, CGPA, Credits

Batch Candidate Verification & Review

Connect Wallet

DH Department Head (L...
Dept Staff

Sign out

Search candidates by name or registration...

STATUS	STUDENT NAME	REGISTRATION NO	CREDITS	CGPA	PASSING YEAR	ACTION
Eligible	Omar Faruk	1803010042	160 Cr	3.95	2026	
Eligible	Tariqul Islam Sunny	1803010043	160 Cr	3.88	2026	
Eligible	SWEETY SUTRADHAR	1916742993	160 Cr	3.92	2026	
Eligible	SAYONTONI DAS NODI	1916742994	160 Cr	3.90	2026	
Eligible	Omar Faruk	1803010042	160 Cr	3.95	2026	

Ready to submit 5 eligible candidate(s) for Main Degree Certificates.

Clear List

Submit 5 Candidates to Admin

Minter Student Request & Approve

CertiChain

Sylhet Engineering College

Connect Wallet

Department Head [...]

Sign out

Department Portal

ACTIVE

DEPARTMENT MANAGEMENT

Dashboard

Student Data Input

Student Records

Certificates & Ledger

Grade Sheets

Student Requests

Storage Status

My Profile

Student Requests

Review and endorse student submissions

Search student, reg, category...

All Pending Approved Rejected

Profiles Grade Sheets Provisional Revocations Leaves Clearance

STUDENT	CORRECTION REQUEST & PREVIEW	STATUS	DOC	ACTIONS
Tariqul Islam Sunny Reg: 1803010044 1803010044@sec.ac.bd +880 1712-345678	Email Address autosp@sec.ac.bd original one is here Profile Photo Update PREVIOUS NEW Student submitted updated profile photo for verification and academic records	PENDING		Verify Diff Reject Endorse
Tariqul Islam Sunny Reg: 1803010042 1803010042@sec.ac.bd +880 1712-345678	PREVIOUS NEW Student submitted updated profile photo for verification and academic records	PENDING	Captured P...	Verify Diff Reject Endorse

Sylhet Engineering College

CertiChain – Academic Credential Registry

Blockchain-based academic certificate issuance, zero-knowledge grade verification, and tamper-proof credential registry for Sylhet Engineering College (Affiliated with SUST).

Polygon Amoy (80002)

IPFS Photo Node Active

PUBLIC VERIFICATION

Verify Certificate

Verify Grade Sheet

Zero-Knowledge Proofs

Student Registration

Student Portal

DOCUMENTATION

Platform Documentation

Verification REST API

Smart Contract Registry

Architecture & Security

SEC GOVERNANCE

About SEC CertiChain

Privacy Policy

Terms of Service

SEC Official Portal

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Privacy Terms Public Verifier

Student Requests

Review and endorse student submissions

Search student, reg, category...

All Pending Approved Rejected

Profiles Grade Sheets Provisional Revocations Leaves Clearance

STUDENT	CORRECTION REQUEST & PREVIEW	STATUS	DOC	ACTIONS
Omar Faruk Reg: 1803010042 1803010042@sec.ac.bd +880 1712-345678	Profile Photo Update PREVIOUS NEW Student submitted updated profile photo for verification and academic records	APPROVED	Captured P...	Verify Diff

Figure 8.4 Minter Portal

8.6 Academic Admin Controller

The Academic Admin Controller is the central authority for governance-sensitive actions. The admin action center organizes approval pipelines so that issuance, grades, profile corrections, download passes, and revocations can be reviewed separately.

The screenshot shows the Super Admin Dashboard for CertiChain. The interface includes a sidebar with navigation options like Admin Portal, Dashboard, Action Center, Student Records, Department Staff, Notices, Audit Trail, Settings, and My Profile. The main content area features a welcome message, a summary of pending actions (4 total: 1 Degree Minting, 1 Revocations, 2 Profile Updates), and a grid of key metrics: Total Issued Certificates (2), Active Faculty & Staff (9), Registered Students (9), Pending Revocations (1 Action), Certificates Issued (2), Active Staff (10), Students Registered (9), and Audit Events (35). A recent activity log at the bottom shows events like 'RevocationMinterEndorsed' and 'RevocationCorrectionRequested'.

Admin Action Center

The screenshot displays the Admin Action Center, a hub for reviewing and approving pending institutional actions. It features a top navigation bar with tabs for Degree Minting, Staff Privileges, Cohort Results, Semester Grades, Profiles & Grades Diff, Download Passes, and Revocations. Below this, a summary of proposals is shown: All Proposals (3), Initial Admin Review (1), In Signatory Committee (1), Ready for Polygon Deployment (0), and Deployed on Amoy (1). The main table lists three proposals for 'Computer Science & Engineering (CSE) Batch' candidates, each with a governance pipeline status (e.g., 'AWAITING DEPT MINTER ENDORSEMENT', 'SIGNATORY PIPELINE (PREPARER)', 'DEPLOYED ON AMOY') and verification actions (Inspect, Approve, Reject, etc.). The bottom section provides information about Sythet Engineering College, including public verification details, documentation links, and governance policies.

Admin Department Staff

Admin Portal Active

ADMINISTRATION

- Dashboard
- Action Center 1
- Student Records
- Department Staff**
- Notices
- Audit Trail
- Settings
- My Profile

Department Minter RBAC Governance
Authorize, monitor, and revoke smart contract minting privileges for departmental faculty and staff.

Search faculty name, email, department, or staff ID... All Departments All Roles All Status Authorize New Staff

Institutional Staff & Faculty Authority Directory
Manage department minters, verification officers, and granular blockchain signing permissions 10 Authorized

FACULTY / OFFICER	DEPARTMENT & ROLE	GRANTED PERMISSIONS	STATUS	ACTIONS
D Department Head (Minter) rejunior@sec.ac.bd	Computer Science & Engineering (CSE) MINTER	MINT_CERTIFICATES INGEST_STUDENTS MANAGE_GRADES PROVISIONAL_MINTING	Active	Permissions Suspend
E Engr. Kazi Mahfuzur Rahman mahfuz.ce@sec.ac.bd	Civil Engineering (CE) MINTER	MINT_CERTIFICATES INGEST_STUDENTS MANAGE_GRADES PROVISIONAL_MINTING	Active	Permissions Suspend
P Prof. Syed Enamul Hoque hoque.eee@sec.ac.bd	Electrical & Electronic Engineering (EEE) MINTER	MINT_CERTIFICATES INGEST_STUDENTS MANAGE_GRADES PROVISIONAL_MINTING	Active	Permissions Suspend
D Dr. Md. Abdur Rahim rahim@sec.ac.bd	Computer Science & Engineering (CSE) MINTER	MINT_CERTIFICATES INGEST_STUDENTS MANAGE_GRADES PROVISIONAL_MINTING	Active	Permissions Suspend
M Md. Al-Amin preparer@sec.ac.bd	Academic Council & Examination Division SIGNATORY	SIGNATORY_ACCESS VIEW_AUDIT	Active	Permissions Suspend
S Shafiqul Islam comparer@sec.ac.bd	Academic Council & Examination Division SIGNATORY	SIGNATORY_ACCESS VIEW_AUDIT	Active	Permissions Suspend
E Engr. Mohammad Selim deputy.controller@sec.ac.bd	Academic Council & Examination Division SIGNATORY	SIGNATORY_ACCESS VIEW_AUDIT	Active	Permissions Suspend

SA Super Admin Admin +

SEC CertChain System

Figure 8.5 Admin Portal

8.7 Signature Portal

- The Signatory Review Portal functions as the institutional clearance and endorsement gateway for academic authorities such as Deans, Tabulation Officers, and Examination Controllers:
 - Granular Role-Based Pipeline:** Enforces sequential multi-tier clearance across designated institutional roles (Preparer, Comparer, and Controller) before a credential reaches the minting phase.
 - Digital Endorsement & Sealing:** Allows authorized signatories to upload, draw, or replace their official digital signature and automatically stamp the institutional seal onto verified provisional and main degree records.
 - Pre-Issuance Academic Audit:** Provides an interface for inspecting 8-semester course ledgers, cumulative CGPA, and candidate dossiers to ensure strict academic validity before on-chain commitment.
 - Traceable Endorsement History:** Maintains an active audit log tracking completed sign-offs, forwarding dates, and reviewer timestamps to uphold administrative accountability.

Academic Credential Signatory Review Portal
PREPARED BY (EXAM OFFICER)

My Digital Signature

Awaiting My Signature (0)
Endorsement History (1)
Student Directory (7)

Completed Endorsement History
 Ledger of verified, signed, and forwarded credentials

CANDIDATE STUDENT	REGISTRATION NO (ID)	DEGREE & TYPE	CGPA / CREDITS	PIPELINE STATUS	VERIFICATION & ACTIONS
CS Computer Science & Engineering (CSE) Session 2022-2023 Cohort Computer Science & Engineering (CSE)	N/A 2022-2023	Bachelor of Science in Computer Scie... PROVISIONAL DEGREE	CGPA: 3.95 160 Credits	PUBLISHED	Inspect Dossier Endorsed

Digital Signature Management

Upload or draw your official digital signature for certificate endorsement.

Active Digital Signature
 This signature will be applied to all certificate endorsements

Replace signature
 Clear

Officer Identity
 Your designation and institutional seal

FULL NAME
 Md. Al-Amin

DESIGNATION / TITLE
 Prepared By (Exam Officer)

INSTITUTIONAL SEAL (FIXED)

Sylhet Engineering College
 Official University Embossed Seal
 Non-editable, date-stamped on approval

Figure : 8.6 Signature Portal

8.8 Certificate Correction Implementation

A central design principle is that a corrected certificate should not silently replace the old certificate. Instead, the system creates a traceable lifecycle. The student starts the request, the department verifies the evidence, and the admin executes the final blockchain action.



Student Grade Sheet


SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY
SVLJET ENGINEERING COLLEGE
Affiliated with SUST - Sylhet-3100, Bangladesh
OFFICIAL ACADEMIC GRADE SHEET (ALL 8 SEMESTERS)


STUDENT NAME Tariqul Islam Sunny	REGISTRATION/PERMANENT ID: 1803010442	ACADEMIC SESSION 2022-2023	DEPARTMENT Computer Science & Engineering (CSE)
FATHER'S NAME Mrs. Rafiqul Islam	MOTHER'S NAME Mrs. Salma Khatun	DATE OF BIRTH 2001-08-22	GRADUATION STATUS Conferred Passures

1st Year 1st Semester (2022) SGPA: 3.89

CODE	COURSE TITLE	CR	GRADE	GP
CSE-1101	Introduction to Computer System...	3	A+	4.00
CSE-1102	Structured Programming Laborat...	1.5	A+	4.00
MATH-1101	Differential & Integral Calculus	3	A	3.75
PHY-1101	Physics I (Mechanics, Waves & O...	3	A	3.75
PHY-1102	Physics Laboratory	1.5	A+	4.00
ENG-1101	Technical English & Communication	3	A+	4.00
EE-1101	Basic Electrical Engineering	3	A+	3.75
EE-1102	Basic Electrical Engineering Lab	1.5	A+	4.00

1st Year 2nd Semester (2022) SGPA: 3.89

CODE	COURSE TITLE	CR	GRADE	GP
CSE-1201	Discrete Mathematics	3	A+	4.00
CSE-1202	Object Oriented Programming	3	A+	4.00
MATH-1201	Linear Algebra & Complex Analysis	3	A	3.75
CHEM-1201	Engineering Chemistry	3	A	3.75
CSE-1203	OOP Lab (Java & C++)	1.5	A+	4.00
CHEM-1202	Chemistry Laboratory	1.5	A+	4.00
CSE-1204	Engineering Workshop & Drafting	2	A+	4.00
BM-1201	Principles of Economics & Socio...	3	A	3.75

2nd Year 1st Semester (2023) SGPA: 3.89

CODE	COURSE TITLE	CR	GRADE	GP
CSE-2101	Data Structures & Algorithms I	3	A+	4.00
CSE-2102	Digital Logic Design	3	A+	3.75
MATH-2101	Calculus II & Differential Equations	3	A	3.75
CSE-2103	Data Structures Lab	1.5	A+	4.00
CSE-2104	Digital Logic Design Lab	3	A+	4.00
CSE-2105	Applied Electronics Lab	1.5	A+	4.00
MATH-2102	Fourier Analysis & Laplace Transf...	3	A	3.75
BM-2101	Society, Ethics & Professionalism	2	A+	4.00

2nd Year 2nd Semester (2023) SGPA: 3.89

CODE	COURSE TITLE	CR	GRADE	GP
CSE-2201	Algorithm Design & Analysis II	3	A+	4.00
CSE-2202	Computer Architecture & Organa...	3	A	3.75
CSE-2203	Database Management Systems	3	A+	4.00
CSE-2204	Database Systems Lab	1.5	A+	4.00
CSE-2205	Numerical Methods in Computing	3	A+	4.00
CSE-2206	Numerical Analysis & Simulation L...	1.5	A+	4.00
STAT-2201	Probability & Statistics for Engin...	3	A	3.75
CSE-2207	Software / Engineering Project I	2	A+	4.00

3rd Year 1st Semester (2024) SGPA: 3.89

CODE	COURSE TITLE	CR	GRADE	GP
CSE-3101	Operating Systems & System Prog...	3	A	3.75
CSE-3102	Data Communication & Computer...	3	A+	4.00
CSE-3103	Software Engineering & Design Pa...	3	A	3.75
CSE-3104	Theory of Computation & Automata	3	A+	4.00
CSE-3105	Operating Systems Lab	1.5	A+	4.00
CSE-3106	Networking & Communication Lab	1.5	A+	4.00
CSE-3107	Engineering Management & Finance	3	A	3.75
CSE-3108	Industrial Training & Field Visit	2	A+	4.00

3rd Year 2nd Semester (2024) SGPA: 3.89

CODE	COURSE TITLE	CR	GRADE	GP
CSE-3201	Microprocessors & Embedded Sys...	3	A+	4.00
CSE-3202	Artificial Intelligence & Neural Net...	3	A+	4.00
CSE-3203	Web Technologies & Cloud Applic...	3	A	3.75
CSE-3204	AI & Machine Learning Lab	1.5	A+	4.00
CSE-3205	Embedded Systems & Robotics Lab	1.5	A+	4.00
CSE-3206	Information System Design & Audit	3	A	3.75
CSE-3207	Technical Seminar & Research Met...	2	A+	4.00
BM-3201	Industrial Law & Safety Manage...	3	A	3.75

4th Year 1st Semester (2025) SGPA: 3.92

CODE	COURSE TITLE	CR	GRADE	GP
CSE-4101	Cryptography & Blockchain Security	3	A+	4.00
CSE-4102	Compiler Design & Formal Langua...	3	A	3.75
CSE-4103	Distributed Systems & Cloud Con...	3	A+	4.00
CSE-4104	Blockchain & Systems Security Lab	1.5	A+	4.00
CSE-4105	Compiler & Systems Software Lab	1.5	A+	4.00
CSE-4106	Advanced Elective I	3	A	3.75
CSE-4107	Senior Design Project / Thesis I	3	A	4.00
NET-4101	Engineering Economics & Project ...	2	A+	4.00

4th Year 2nd Semester (2025) SGPA: 3.94

CODE	COURSE TITLE	CR	GRADE	GP
CSE-4201	Decentralized Applications & Smar...	3	A+	4.00
CSE-4202	Deep Learning & Big Data Analytics	3	A+	4.00
CSE-4203	Engineering Ethics & Intellectual Pr...	2	A+	4.00
CSE-4204	Senior Capstone Thesis & Defense	4	A+	4.00
CSE-4205	Smart Contracts & Big Data Lab	1.5	A+	4.00
CSE-4206	Comprehensive Academic Viva Vo...	2	A+	4.00
CSE-4207	Advanced Elective II (Cyber Phys...	3	A	3.75
BM-4201	Entrepreneurship & Innovation Str...	1.5	A+	4.00

Total Completed Credits: 150.0 Credits
Cumulative CGPA: 3.95 / 4.00 (First Class with Honours)



Official Verification Provenance
Scan QR or visit <http://localhost:3000/verify>


Head of Department
Computer Science & Engineering


Controller of Examinations
SUST, Sylhet

Figure 8.7 Certificate Portal

8.9 Zero-Knowledge Proof Module

The proof module is designed around user-friendly presets rather than requiring students to construct cryptographic circuits manually. Each preset represents a clear statement. The resulting proof can be shared with an employer or institution without exposing the protected underlying values.

Preset	Statement Demonstrated	Protected Information
Minimum CGPA	CGPA is greater than or equal to a threshold.	Exact CGPA
Degree Completion	Required credits are completed and graduation status is true.	Full grade history
Department	Student belongs to a specified department.	Personal identifiers
Passing Year	Graduation occurred within a specified year condition.	Unnecessary academic details

8.10 Client-Side Biometric Module

The biometric module follows a three-step browser workflow: retrieve the official reference photo, capture a live image, and compare facial descriptors. The implementation described in the project summary uses TensorFlow.js and @vladmandic/face-api with a 128-dimensional descriptor and distance-based comparison.

The result is intended to support manual identity checking rather than replace certificate verification. A certificate can be genuine while being presented by someone other than the named student. The biometric step addresses this separate identity problem.

8.11 Notifications and Audit

Notifications make the multi-tier workflow understandable to users. For example, a student should know when a correction request has been received, endorsed, approved, rejected, or re-minted. Audit logs provide the institutional record needed to reconstruct important security and governance events.

Chapter 10

Certificate Lifecycle, Security and Governance

Security in CertiChain is implemented as a layered property rather than as a single feature. Blockchain immutability protects the integrity anchor, application authorization protects user actions, IPFS separates large files, zero-knowledge proofs reduce unnecessary disclosure, and biometric checking adds a practical identity signal.

10.1 Authentication

Students use a combination of application identity and wallet binding. The registration workflow includes an OTP step so that possession of the student's institutional email is linked to the onboarding process. Web3 actions use MetaMask to sign transactions without exposing private keys to the application.

10.2 Authorization

Authorization occurs at both the application and smart contract levels. The application determines what interface and business functions a role can access, while the smart contract independently restricts sensitive on-chain operations to authorized issuer wallets or the contract owner.

10.3 Certificate States

State	Meaning	Verification Result
Active	Certificate exists and has not been revoked.	Valid, subject to integrity checks.
Revoked	Certificate has been invalidated by the governance process.	Invalid / revoked.
Corrected	A replacement certificate has been issued after a correction.	Verify the new certificate identity.
Pending Request	A correction/revocation request is being reviewed.	Existing certificate state remains unchanged until approval.

10.4 Revocation and Burn

Revocation is necessary because real academic records sometimes contain mistakes, are compromised, or need correction. CertiChain treats revocation as a state transition rather than deletion. This preserves the audit trail while preventing the old credential from being presented as active.

10.5 Atomic Burn and Re-Mint

The current project specification describes an atomic governance flow in which the old certificate is marked revoked, database values are corrected, a new hash is calculated, and a corrected certificate is registered. This design reduces the risk of ending with a corrected database record but an unchanged blockchain credential.

10.6 Privacy Model

CertiChain follows a data minimization principle. The public verifier should receive enough information to determine credential validity, but not automatically receive the student's complete academic profile. Full documents can be shared when the student chooses to provide them, while zk-proofs allow some common verification questions to be answered without disclosing the underlying values.

10.7 Threat Model

Threat	Potential Impact	Mitigation
Fake certificate	Fraudulent qualification	Blockchain commitment and verification portal.
Database alteration	Incorrect institutional data	Blockchain integrity anchor and audit trail.
Unauthorized issuance	Invalid credential creation	Authorized issuer mapping + admin governance.
Certificate sharing by wrong person	Identity fraud	Client-side face comparison + student-controlled sharing.
Privacy leakage	Exposure of PII/academic data	Off-chain separation + zk-proof selective disclosure.
Malicious correction	Wrong academic record	Student evidence + minter endorsement + admin approval.
Duplicate issuance	Conflicting credentials	Unique certificate hash and institutional pre-registration checks.
Contract misuse	On-chain state corruption	Access modifiers, validation, testing, and limited contract scope.

10.8 Security Limitations

No security architecture can remove every risk. A compromised student's email account can weaken OTP-based onboarding; a compromised wallet can authorize actions available to that wallet; an inaccurate or outdated reference photograph can reduce biometric usefulness; and IPFS availability depends on continued pinning/provider support. These limitations are important because the system should be presented as a layered security architecture rather than an absolute guarantee.

Chapter 11

Testing and Evaluation

Testing is necessary because the project combines several independent technologies. A certificate may be generated by the frontend, stored in the database and IPFS, registered on Polygon, and later verified through a public portal. A failure in any one layer can affect the complete user experience.

11.1 Testing Strategy

Testing Level	Focus
Unit Testing	Smart contract functions, validation logic, hashing, calculation utilities.
Integration Testing	Frontend ↔ database, frontend ↔ IPFS, frontend ↔ blockchain.
Workflow Testing	Registration, issuance, verification, correction, revocation, re-mint.
Security Testing	Authorization, invalid input, unauthorized issuer attempts, revoked credentials.
Performance Testing	Verification response, blockchain calls, UI responsiveness, proof generation.
Usability Testing	Clarity of student and verifier workflows.

11.2 Smart Contract Test Cases

Test Case	Expected Result
Issue with authorized issuer	Certificate record created successfully.
Issue with unauthorized wallet	Transaction rejected by access control.
Issue duplicate hash	Transaction rejected.
Verify existing certificate	Correct existence, issuer, timestamp, and status returned.
Verify unknown hash	Non-existence returned.
Revoke active certificate	Status changes to revoked and count increments.
Revoke already revoked certificate	Transaction rejected.
Authorize issuer by owner	Issuer becomes authorized.
Authorize issuer by non-owner	Transaction rejected.

11.3 Frontend and Workflow Tests

Scenario	Validation
Student registration	Valid pre-registration + correct OTP leads to account activation.
Wrong OTP	Registration does not complete.
Certificate download	Correct credential is generated and linked to verification information.
QR verification	QR resolves to the expected credential verification route.
Revoked certificate	Public portal reports invalid/revoked status.
Correction request	Request progresses through expected role states.
Admin rejection	Rejection reason is recorded and student is notified.
ZK proof	Valid proof passes; malformed or inconsistent proof fails.
Face comparison	Reference and live image produce a similarity result without server-side biometric storage.

11.4 Performance Considerations

The project summary describes the verification goal as near-instant or sub-second for common operations, while the pre-defense materials describe typical verification in a few seconds depending on the method. The practical latency depends on network conditions, RPC response, IPFS retrieval, browser processing, and whether a proof or biometric comparison is performed. Therefore, performance should be evaluated as a range rather than a single guaranteed number.

The gas optimization design is expected to reduce persistent storage requirements substantially because the contract stores only compact commitments. The Merkle-tree strategy is particularly relevant for future graduating cohorts, where hundreds of certificates may otherwise require repeated transaction overhead.

11.5 Evaluation Criteria

Criterion	Success Condition
Integrity	Any change to standardized credential data results in a commitment mismatch.
Authenticity	Only authorized issuers can create valid on-chain credential records.
Revocation	A revoked credential is clearly rejected by verification.
Privacy	ZK verification can confirm supported claims without revealing protected values.
Usability	Public verification can be performed without wallet registration.
Governance	Sensitive corrections require the defined multi-tier approval path.
Scalability	Large files remain off-chain and the on-chain state remains compact.
Auditability	Important credential lifecycle actions are traceable through logs and transaction records.

11.6 Observed Project Outcomes

The current implementation and specification demonstrate the feasibility of integrating the major project components into one platform. The smart contract provides the integrity anchor, the database supports operational workflows, IPFS supports document storage, and the four portals provide role-specific interaction. The project also establishes a clear path for privacy-preserving verification and institutional governance.

At the same time, some features require further production-grade validation before institutional deployment. In particular, zk-proof circuits need formal benchmark and security evaluation, biometric accuracy needs testing across diverse real-world conditions, and the full deployment should undergo independent security review.

Chapter 12

Discussion, Research Contribution and Novelty

This chapter discusses what the project contributes beyond a conventional blockchain certificate prototype. The most important result is architectural: CertiChain treats credential management as a complete lifecycle rather than as a single issuance transaction.

12.1 What Has Been Developed

- A four-actor credential ecosystem with Public Verifier, Student, Department Minter, and Academic Admin Controller.
- A compact CertificateRegistry smart contract on Polygon Amoy.
- A deterministic Keccak-256 credential commitment model.
- Hybrid storage using Polygon, IPFS/Pinata, and Prisma/SQLite.
- Student certificate, gradesheet, transcript, request, transfer/MOI, and proof workflows.
- Department-level academic review and endorsement workflows.
- Central administrative approval and credential lifecycle controls.
- Public verification by serial, registration, hash, QR, and camera.
- zk-SNARK-oriented selective qualification verification.
- Client-side face comparison.
- Revocation, burn, correction, and re-minting workflow.
- Audit logs and role-targeted notifications.

12.2 Originality

The originality of the project is primarily in the integration and governance model. Rather than presenting blockchain as a replacement database, the design uses it as a compact trust anchor. Rather than treating privacy as an optional feature, the system makes selective disclosure part of the verification workflow. Rather than treating certificate correction as a database edit, the system represents it as a governed state transition.

12.3 Novelty

Contribution Area	Novel Aspect in CertiChain
Gas optimization	32-byte credential commitments combined with IPFS storage and Merkle-oriented batch strategy.
Privacy	Practical proof presets tied to common academic verification questions.
Governance	Student-originated request, departmental endorsement, and central administrative execution.
Identity	Optional client-side biometric comparison in the public verification flow.
Lifecycle	Revocation/burn plus deterministic corrected re-mint rather than silent replacement.
Institutional usability	5 role-specific portals around one shared credential lifecycle.

12.4 Significance to the Academic Field

The project demonstrates how modern Web3 mechanisms can be adapted to an institutional process that has traditionally been paper- and database-oriented. The work is relevant to academic administration because it connects technical properties such as immutability and zero-knowledge proofs with practical requirements such as correction, approval, student support, and public verification.

12.5 Significance to Bangladesh

For institutions operating in Bangladesh, a lightweight and modular credential platform could reduce dependence on manual verification and make it easier for graduates to present credentials internationally. The use of Polygon and off-chain storage also reflects the practical need to control infrastructure and transaction costs. However, institutional adoption would require policy decisions, legal review, data protection practices, and integration with existing academic systems.

12.6 Development Challenges

The development of the project requires understanding several domains at once. Smart contracts introduce a different programming model because deployed state is persistent and transaction execution is externally observable. Wallet integration introduces network, account, and transaction states that ordinary web applications do not have. IPFS adds content-addressed storage semantics, while zero-knowledge proofs require cryptographic reasoning beyond standard frontend development.

Another challenge is balancing privacy with verification. Showing too little information may make manual verification difficult, while showing too much can defeat the purpose of privacy-preserving credentials. The final design therefore separates normal certificate verification from claim-only zk verification and treats biometric comparison as an additional identity check.

12.7 Research Contribution Summary

Requirement from Defense Guideline	How CertiChain Addresses It
Research background/problem	Academic forgery, manual verification, privacy, storage, and governance challenges are defined.
Research questions	Five research questions guide the system design and evaluation.
Objectives	Security, privacy, gas efficiency, governance, verification, and usability objectives are specified.
Related work/research gap	Blockchain credentials, IPFS, ZKP, and institutional limitations are reviewed.
Methodology	Structured research and iterative development methodology is described.
Experimental setup/implementation	Technology stack, smart contract, database, portals, IPFS, ZK, and biometric components are documented.
Results/discussion	Implementation outcomes, evaluation criteria, limitations, and practical implications are discussed.
Contribution/originality/novelty	Hybrid trust architecture, multi-tier governance, selective proofs, and lifecycle management are highlighted.
Conclusion/future work	The final chapters summarize findings and future expansion paths.

Chapter 13

Future Work

Although the current platform provides a broad set of features, several improvements are necessary before it can be considered a production institutional service. Future work should prioritize interoperability, formal security validation, scalability, and long-term operational governance.

1. Per semester grade sheet
2. Mobile Application
3. Emergency Leave and Credit Transfer
4. Result generation
5. Clearance request manage

Chapter 14

Conclusion

CertiChain demonstrates a practical architecture for modern academic credential management in which blockchain is used as a compact integrity and governance layer rather than as a replacement for every other storage system. The project addresses the weaknesses of traditional certificate workflows by combining cryptographic commitments, decentralized file storage, public verification, privacy-preserving proofs, role-based governance, and client-side identity checking.

The use of a 32-byte Keccak-256 certificate commitment keeps the on-chain record small. IPFS is used for large documents and photographs, while Prisma and SQLite support operational records that require normal application queries and updates. This separation makes the architecture more practical than a design that stores complete academic documents directly on the blockchain.

A central strength of the system is its five-portal institutional architecture designed to manage the full credential lifecycle. The Student Portal enables two-factor OTP registration, certificate and grade sheet access, document vault storage, and request submissions for record corrections, duplicate reissues, and MOI or sealed envelope services. The Department Minter Portal equips departmental officers with bulk CSV/XLSX ingestion, session-based student dossiers, batch and provisional certificate issuance, single-student transcript building, and student request resolution. The Signatory Portal introduces an isolated review queue for executive document signing and endorsement tracking. The Academic Admin Controller Portal provides central oversight for cohort degree minting, staff authorization, result finalization, system-wide notices, comprehensive multi-category audit trails, and atomic credential revocation with on-chain burning and re-minting. Finally, the Public Verification Portal enables registration-free authentication via certificate hash, dynamic QR code scanning, file upload, zk-SNARK claim validation, and client-side biometric matching.

Privacy is treated as a first-class requirement. Through zk-SNARK-oriented proof presets, a

student can prove selected academic claims without exposing an entire transcript. The client-side biometric component further addresses a different problem: the difference between proving that a credential is genuine and proving that the person presenting it is the student named on the credential.

Overall, the project provides a strong foundation for a decentralized academic credential ecosystem at Sylhet Engineering College. Its long-term value will depend on formal security validation, production deployment planning, interoperability, institutional policy, and integration with existing academic systems. Nevertheless, the current design demonstrates that blockchain, decentralized storage, zero-knowledge proofs, edge biometrics, and multi-tier five-portal governance can be combined into a coherent and human-centered credential platform.

References

- [1] Watane, H.N., Patgiri, R. and Singha, M.F., 2026. Hy-Cert: A Hybrid Certificate Verification System Using Blockchain for Secure Communication. *Concurrency and Computation: Practice and Experience*, 38(6), p.e70655.
- [2] Deepika, S., Durga Devi, M. and Dharani, M., 2026. Block-Chain Based Certificate Verification System. *International Journal of Engineering & Extended Technologies Research (IJEETR)*, 8(2), pp.3706-3712.
- [3] Priyadarshini, R., Pandey, R., Ankit, K.C., Bhandari, D., Khadka, B., Barik, R.K. and Saikia, M.J., 2025. A faster, integrated, and trusted certificate authentication and issuer validation system based on blockchain. *IEEE Access*, 13, pp.27037-27049.
- [4] Sree, T.R., 2025. Decentralized certificate issuance and verification system using Ethereum blockchain technology. *Journal of Network and Computer Applications*, 242, p.104190.
- [5] Kadam, P.A., Bhushan, S., Rishiwal, V., Yadav, M., Yadav, P. and Agarwal, U., 2025. ValidCertify: an effective blockchain-based domain certificate authentication and verification scheme. *Peer-to Peer Networking and Applications*, 18(4), p.177.
- [6] Rustemi, A., Dalipi, F., Atanasovski, V. and Risteski, A., 2023. A systematic literature review on blockchain-based systems for academic certificate verification. *Ieee Access*, 11, pp.64679-64696.
- [7] Said, S.H., Dida, M.A., Kosia, E.M. and Sinde, R.S., 2023. A blockchain-based conceptual model to address educational certificate verification challenges in Tanzania. *Engineering, Technology & Applied Science Research*, 13(5), pp.11691-11704.
- [8] Jaafar, R.A., Alsaad, S.N. and Al-Kabi, M.N., 2024. Educational certificate verification system: Enhancing security and authenticity using ethereum blockchain and ipfs. *Al-Mustansiriyah Journal of Science*, 35(1), pp.78-87.
- [9] Noorhizama, N.K., Abdullah, Z., Kasim, S., Hamid, I.R.A. and Isa, M.A.M., 2023. Verification of ph. d. certificate using qr code on blockchain ethereum. *JOIV: International Journal on Informatics Visualization*, 7(3), pp.716-725.
- [10] Rani, P., Sachan, R.K. and Kukreja, S., 2024. Educert-chain: a secure and notarized educational certificate authentication and verification system using permissioned blockchain. *Cluster Computing*, 27(7), pp.10169-10196.

Appendix A: Main Portal Responsibilities

Portal	Route Family	Key Functions
Public Verifier	/verify	Serial/registration/hash/QR/camera verification, zk-proof verification, optional face comparison.
Student Portal	/student/*	Registration, certificate, gradesheet, transcript, zk-proof, transfer/MOI, requests, revocation.
Department Minter	/minter/*	Whitelist, batch mint preparation, grades, request review, endorsement, IPFS telemetry.
Academic Admin	/admin/*	Approvals, minter authorization, governance, burn/re-mint, branding, notices, audit.

Appendix B: Certificate Hash Specification

The canonical certificate commitment is generated from the standardized sequence: student name, roll number, registration number, department, degree, CGPA, and passing year. Fields are separated by a pipe character before Keccak-256 hashing. The exact canonicalization rules should be kept identical across issuance and verification.

```
canonical = studentName + "|" + rollNo + "|" + regNo + "|" +  
            department + "|" + degree + "|" + cgpa + "|" +  
            passingYear
```

```
certHash = keccak256(bytes(canonical))
```

Appendix C: Sample Verification Decision Logic

1. Receive certificate identifier.
2. Resolve identifier to certHash.
3. Query CertificateRegistry.verifyCertificate(certHash).
4. If exists == false -> INVALID.
5. If isRevoked == true -> INVALID / REVOKED.
6. Confirm issuer is an authorized institutional issuer.
7. If document preview is requested, retrieve IPFS content.
8. Recompute expected hash from canonical metadata when applicable.
9. If zk-proof is supplied, verify the proof separately.
10. If face verification is requested, run comparison client-side.
11. Return only the minimum information needed for the selected verification mode.

Appendix D: Defense Presentation Alignment

The supplied departmental defense guideline requires a concise presentation covering title, background/problem, research questions or hypothesis, objectives, related work/research gap, methodology, implementation, results and discussion, research contribution/originality/novelty, conclusion, and future work. This report has been organized so that these elements can be directly converted into presentation slides.

- Title and motivation: Chapter 1.
- Research questions and objectives: Sections 1.3–1.5.
- Related work and research gap: Chapter 3.
- Methodology: Chapter 6.
- Implementation: Chapter 7.
- Results and evaluation: Chapter 9.
- Contribution, originality, and novelty: Chapter 10.
- Conclusion: Chapter 12.
- Future work: Chapter 11.